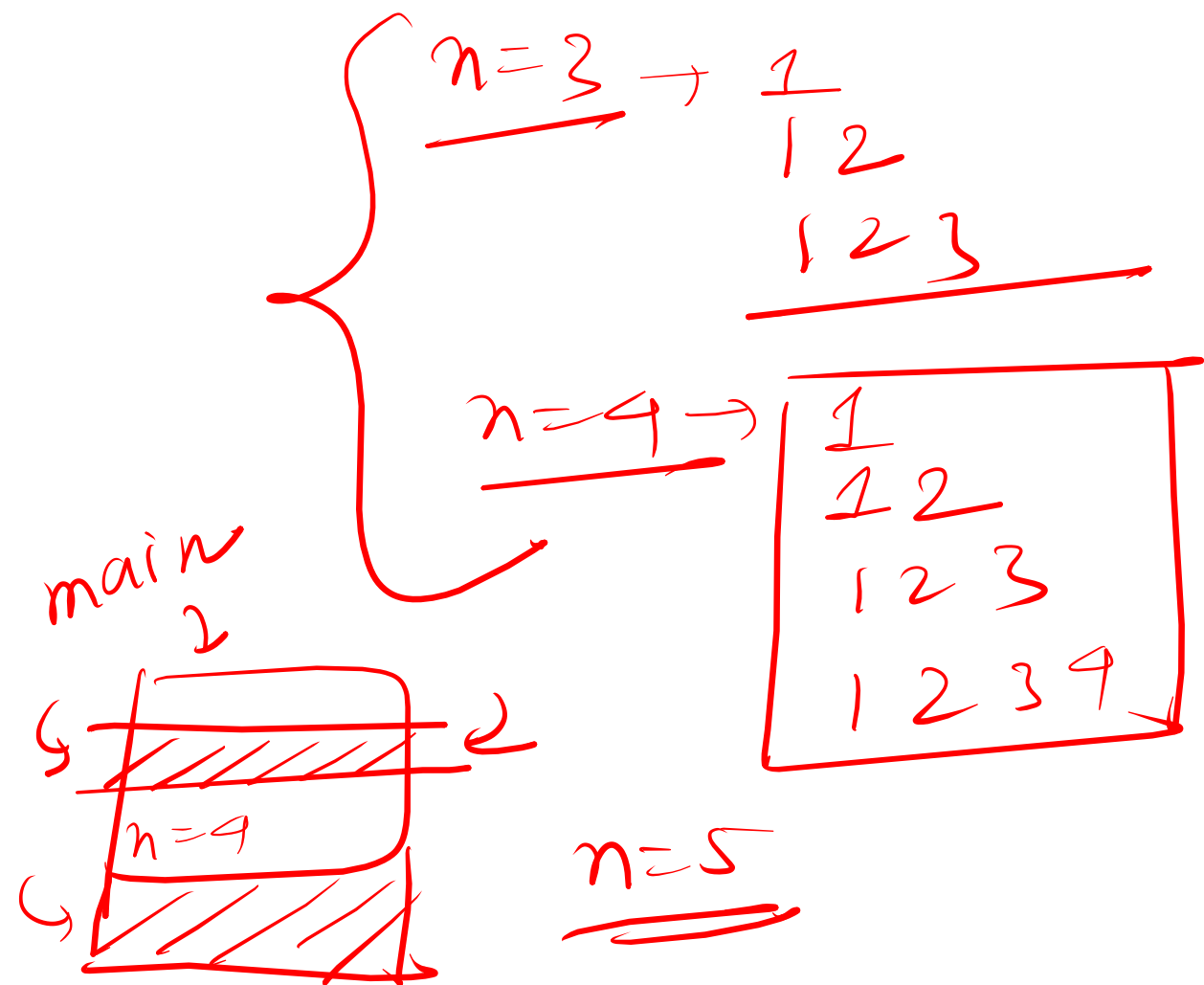


$$\begin{array}{l}
 x = \cancel{1} \cdot 2 \\
 \hookrightarrow (\cancel{1+1}) \rightarrow 2 \\
 \hline
 x + \cancel{1} \quad 2 \\
 \hline
 x = \cancel{2} \rightarrow 3
 \end{array}$$

$$\begin{array}{l}
 \textcircled{+ - x -} \quad \textcircled{x} = \cancel{2} \textcircled{3} \\
 \textcircled{3} \\
 \hline
 x = \cancel{3} / \quad \cancel{x + 1} \\
 \quad \quad \quad 3
 \end{array}$$

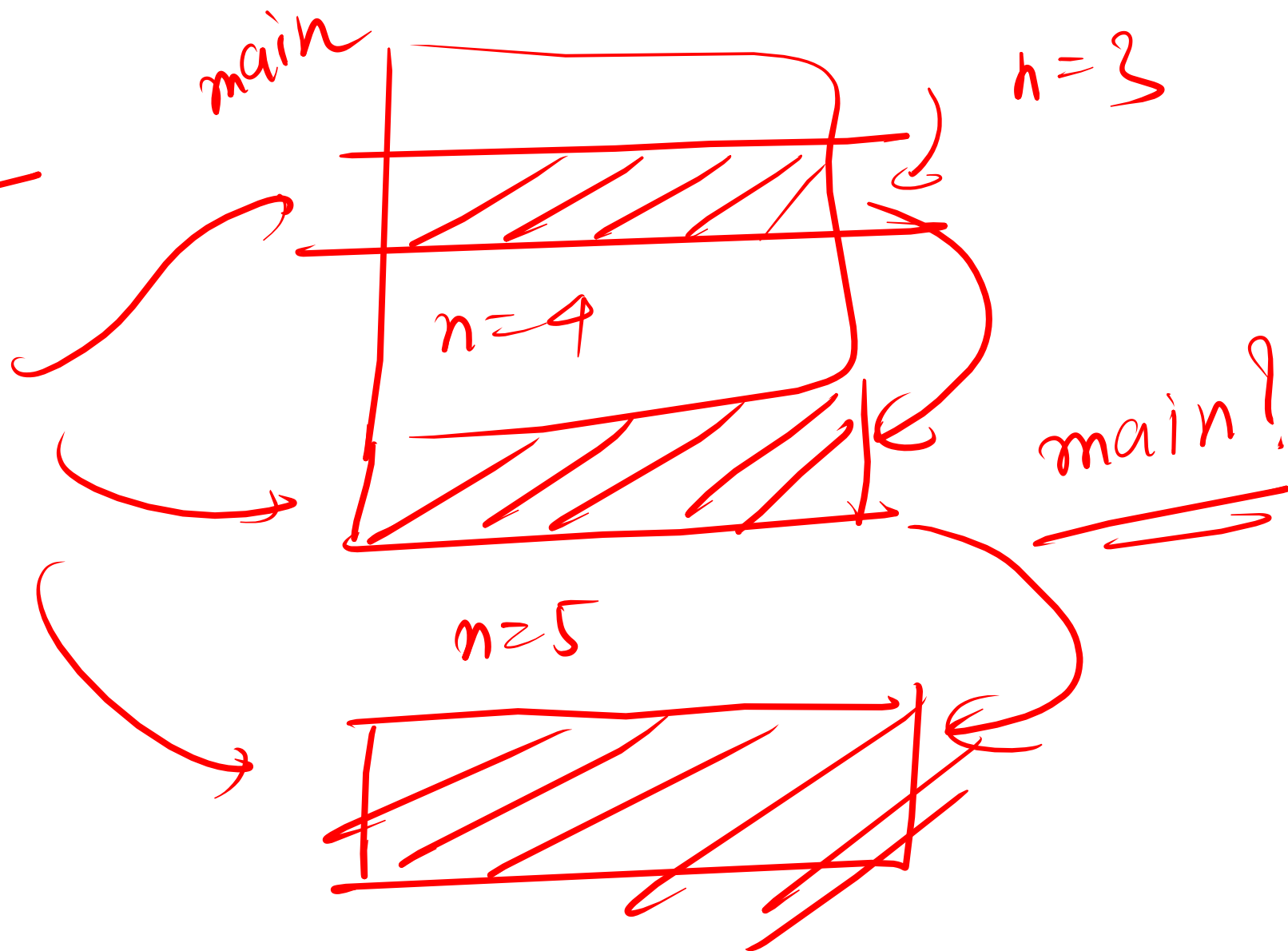
$n=3$
 function
 $\text{for}(i=1; i \leq n; i++) \{$
 $\text{for}(j=1; j \leq i; j++) \{$
 $\text{print}(j + " ");$
 $\}$
 $\text{print}("\n");$
 $\}$

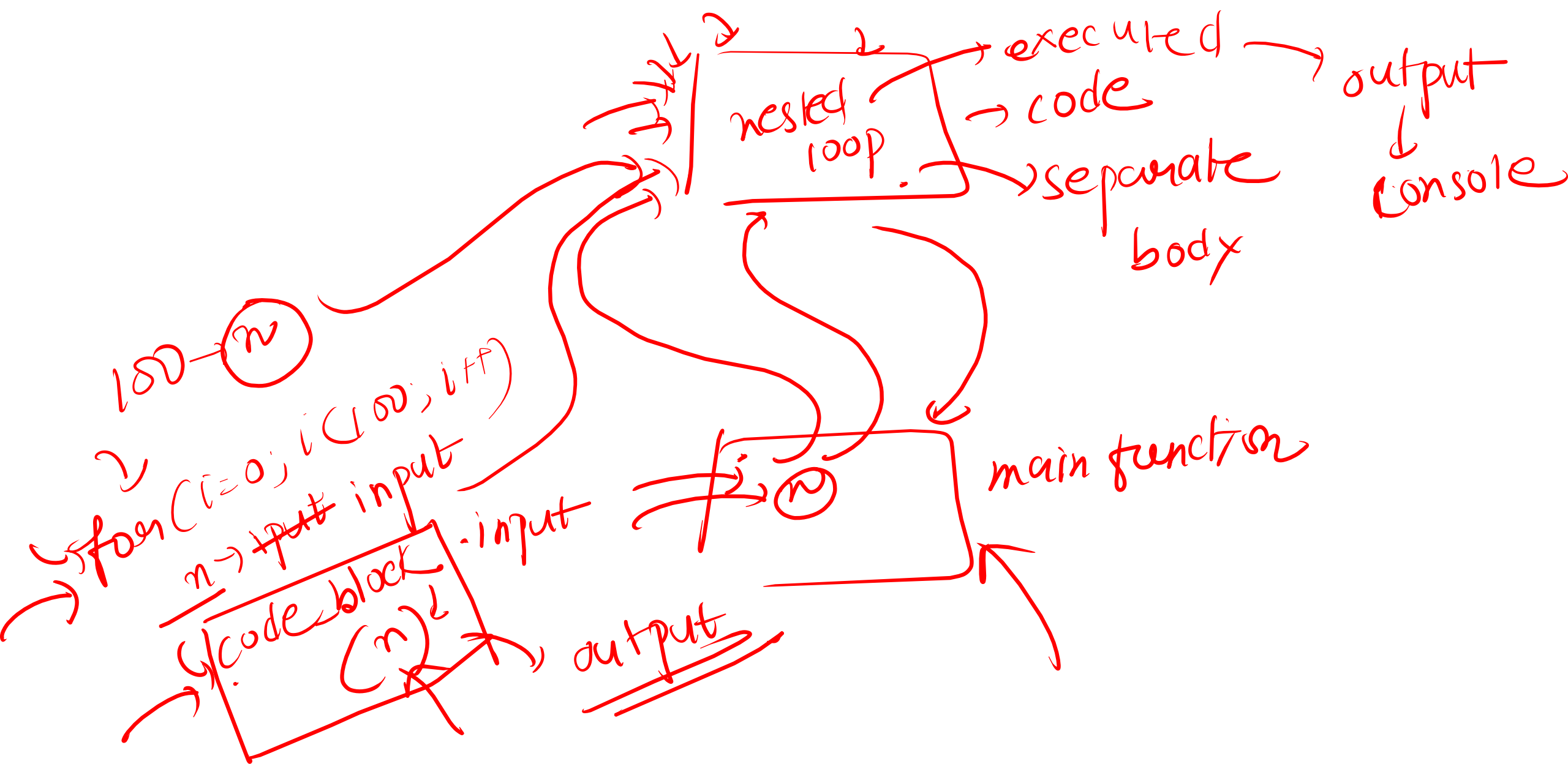


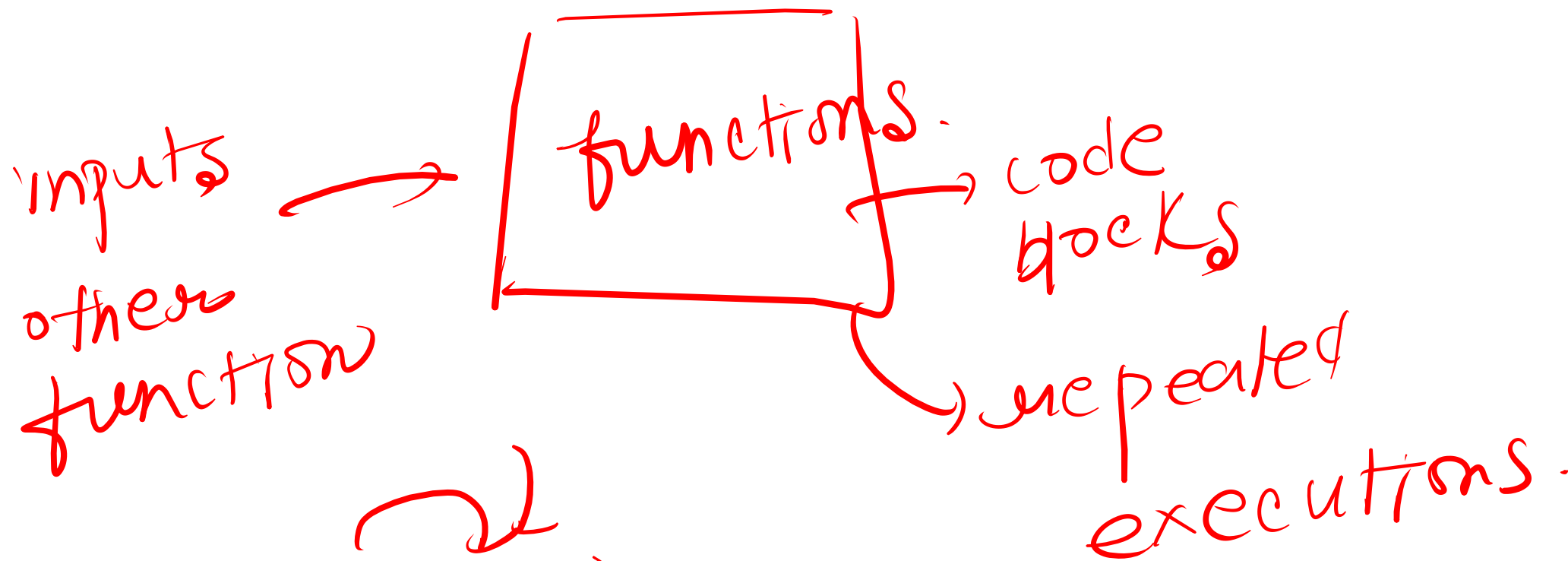
100 test cases.

$n=1$
:
10

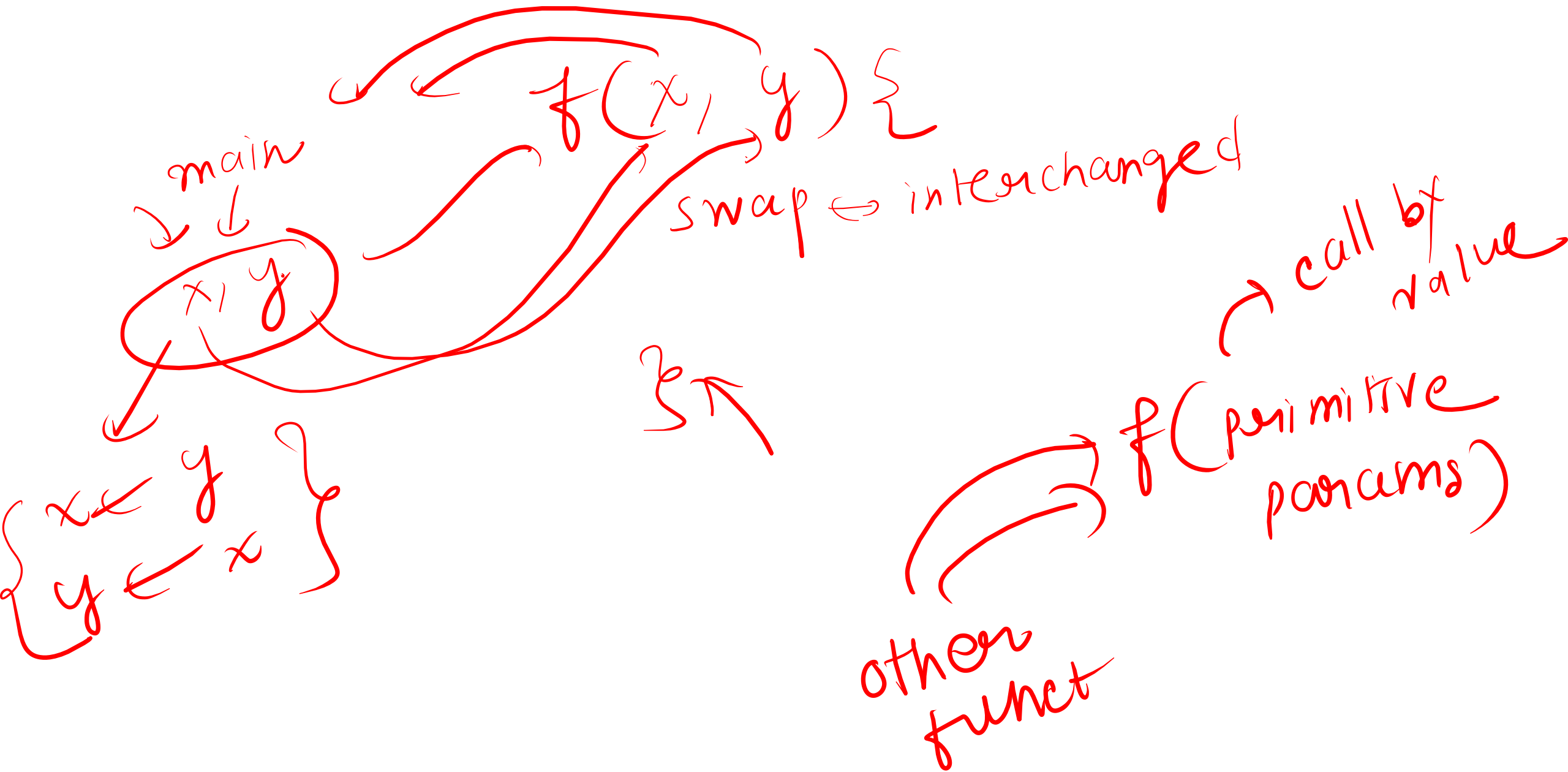
n^2







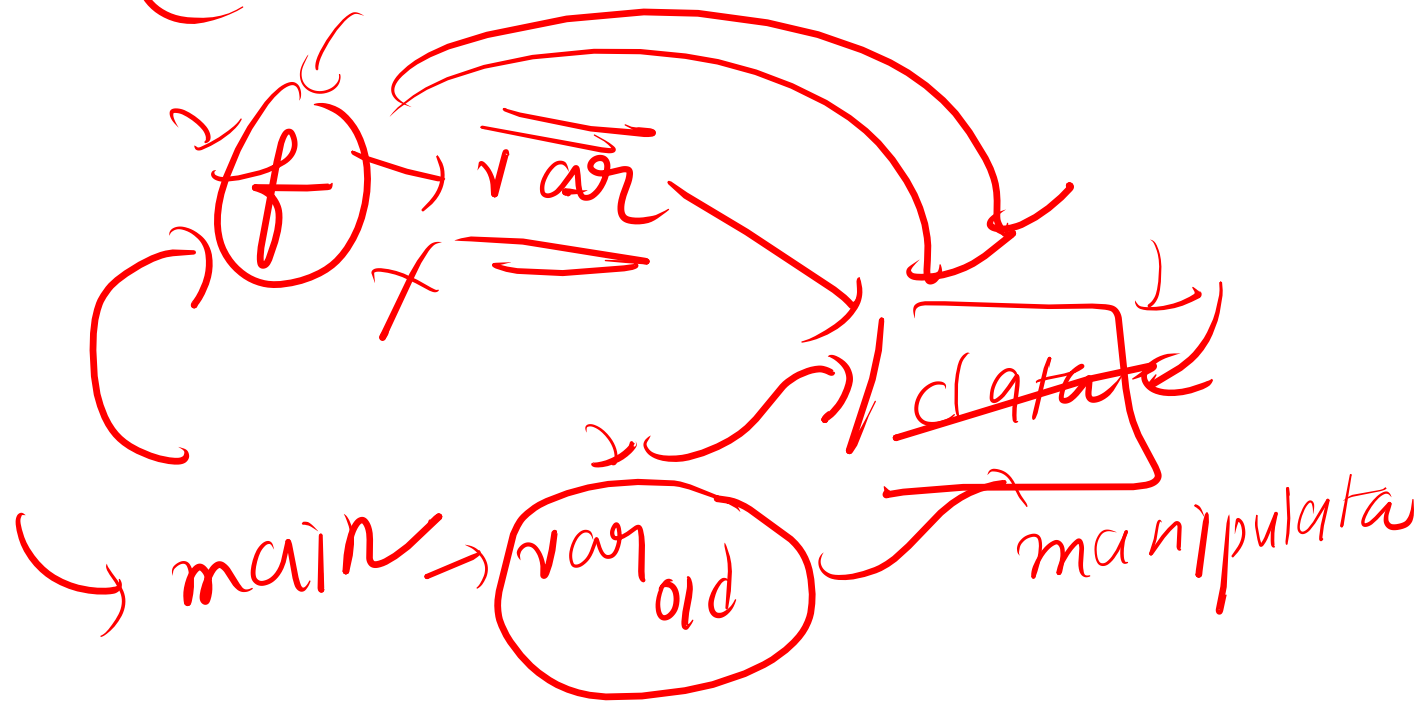
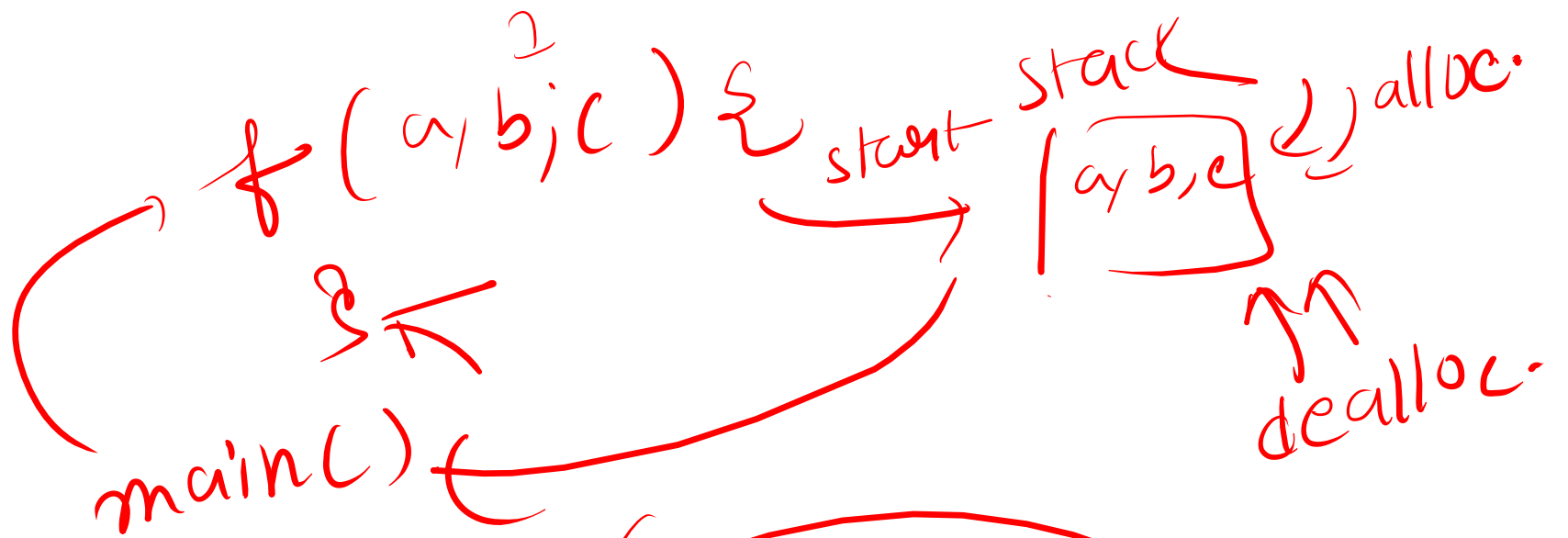
$f(\underline{n})$



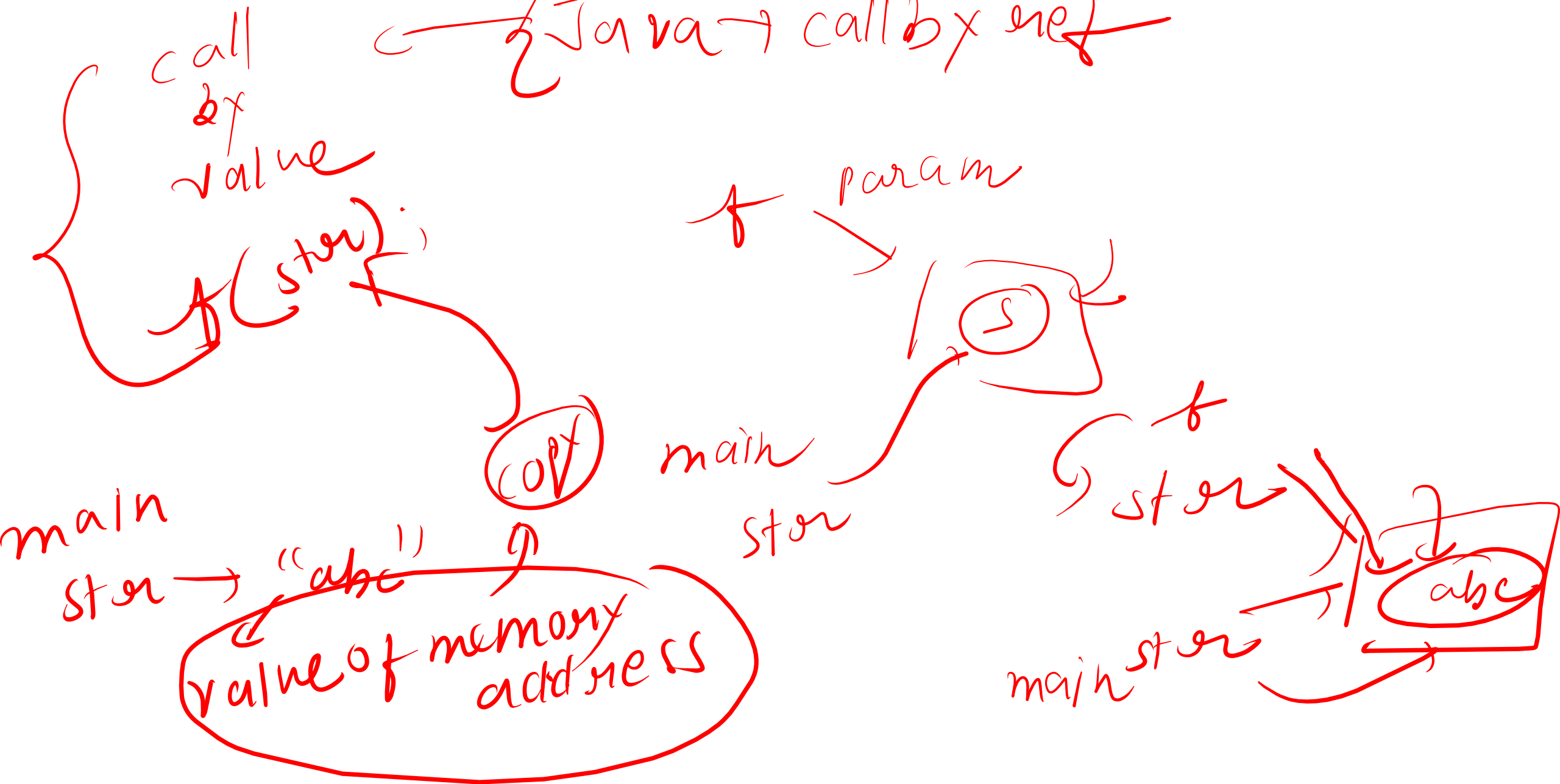
$f(\text{object})$

{ call by ref? }?

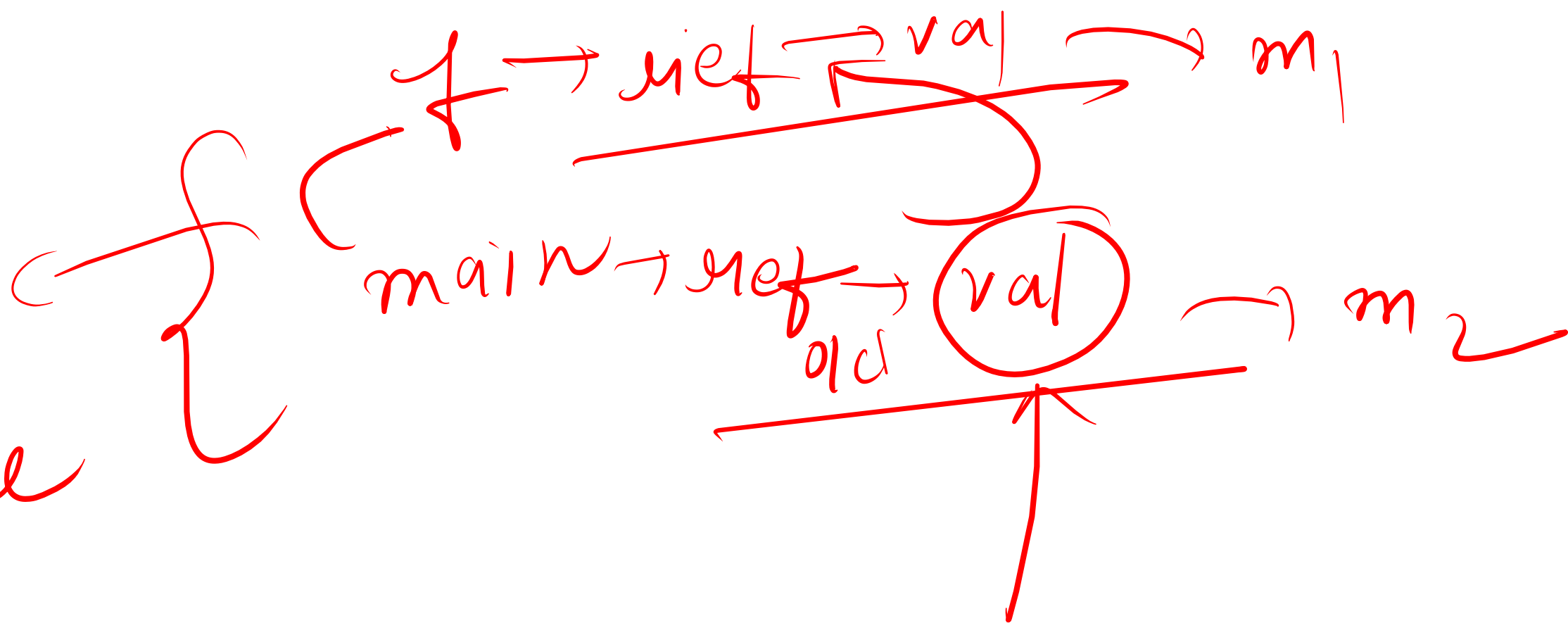
$f \rightarrow \text{ref}$ $\rightarrow$ same
 $\text{main} \rightarrow \text{ref}$ $\rightarrow$ old



Java call by ref

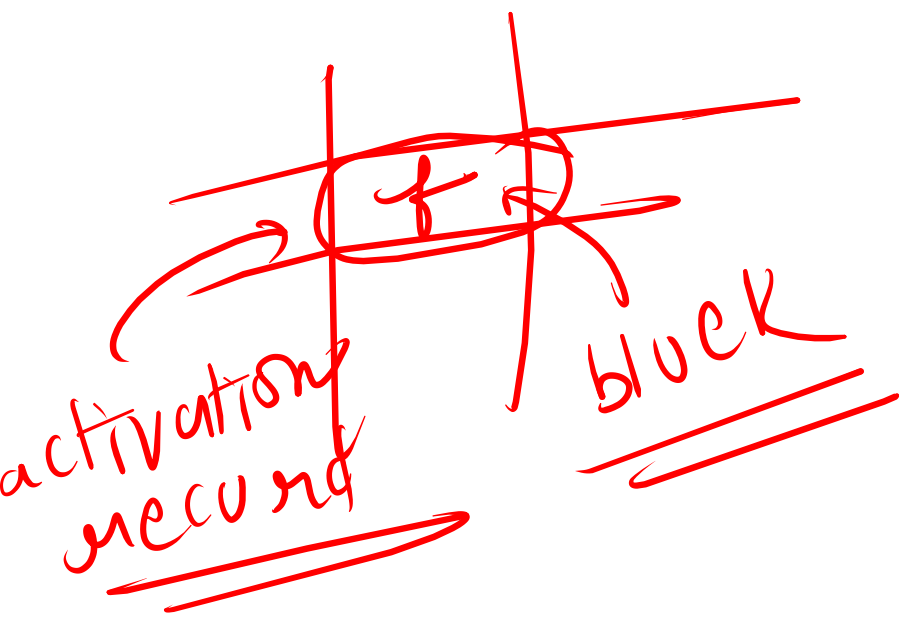


call
of
value



activation record

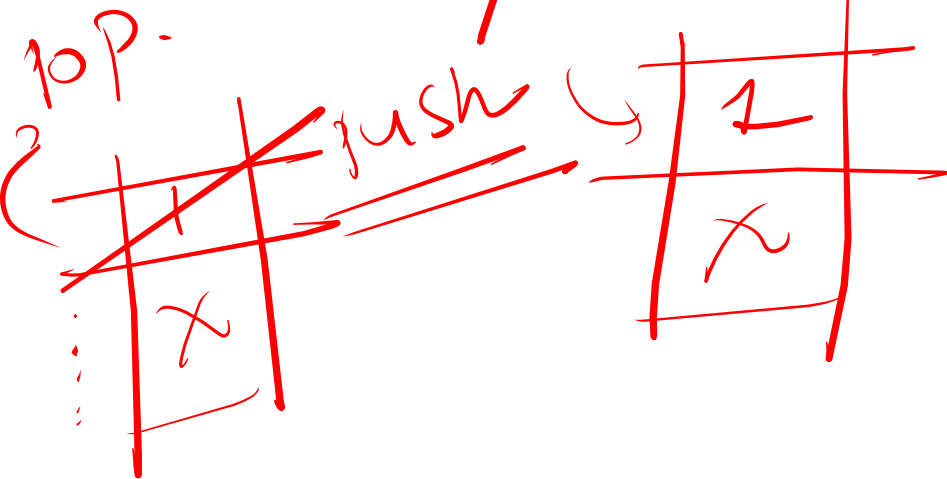
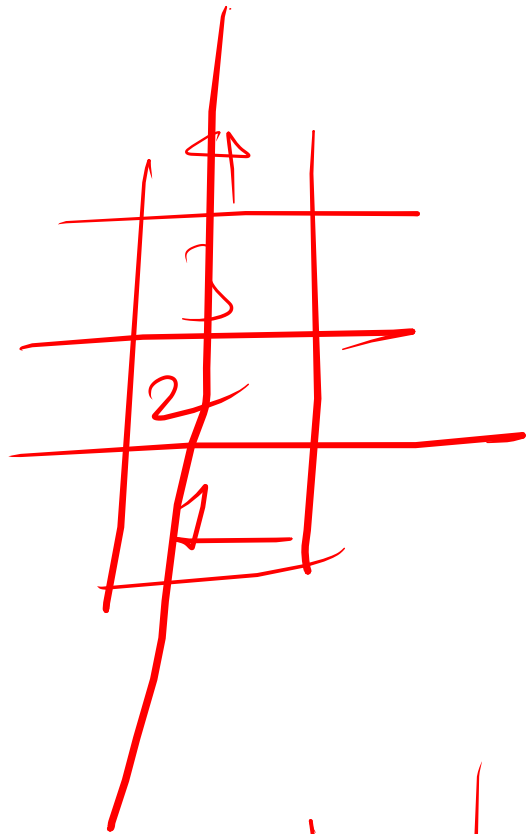
block



$f(\text{object}) \Rightarrow \left\{ \begin{array}{l} \text{assume} \\ \text{call by ref} \end{array} \right\}$

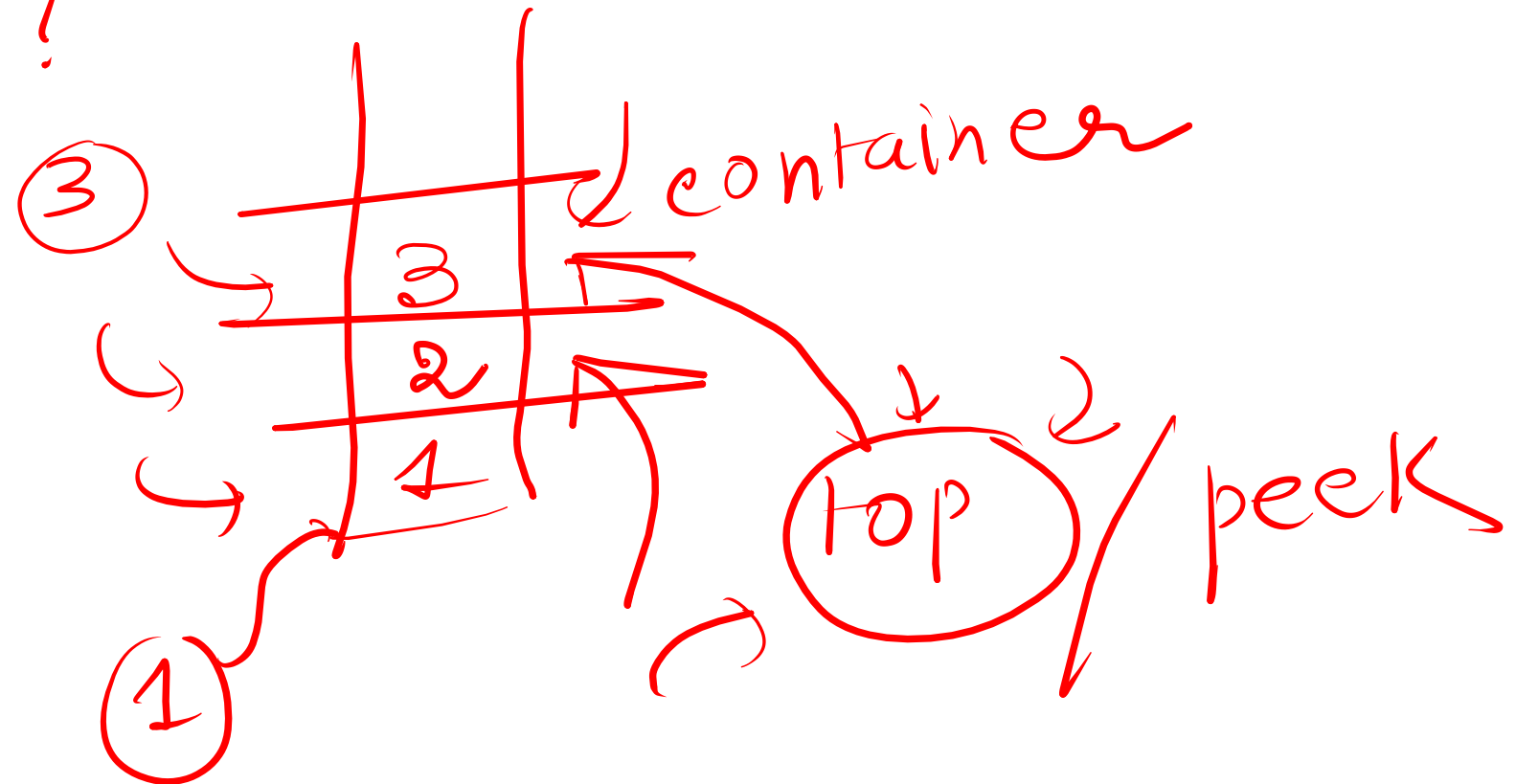
$\swarrow$

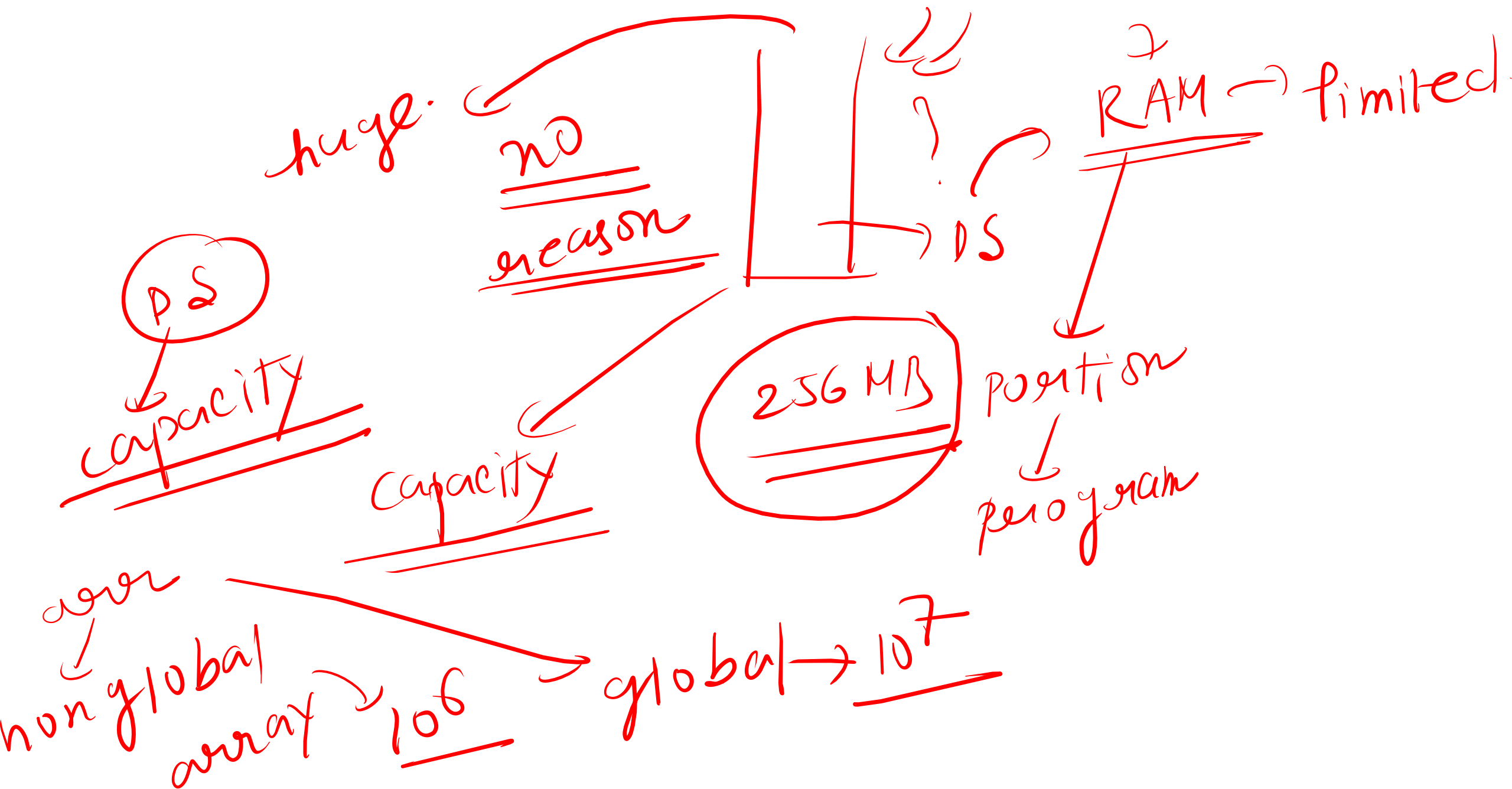
$\left\{ \begin{array}{l} \text{multiple} \\ \text{task} \end{array} \right\}$

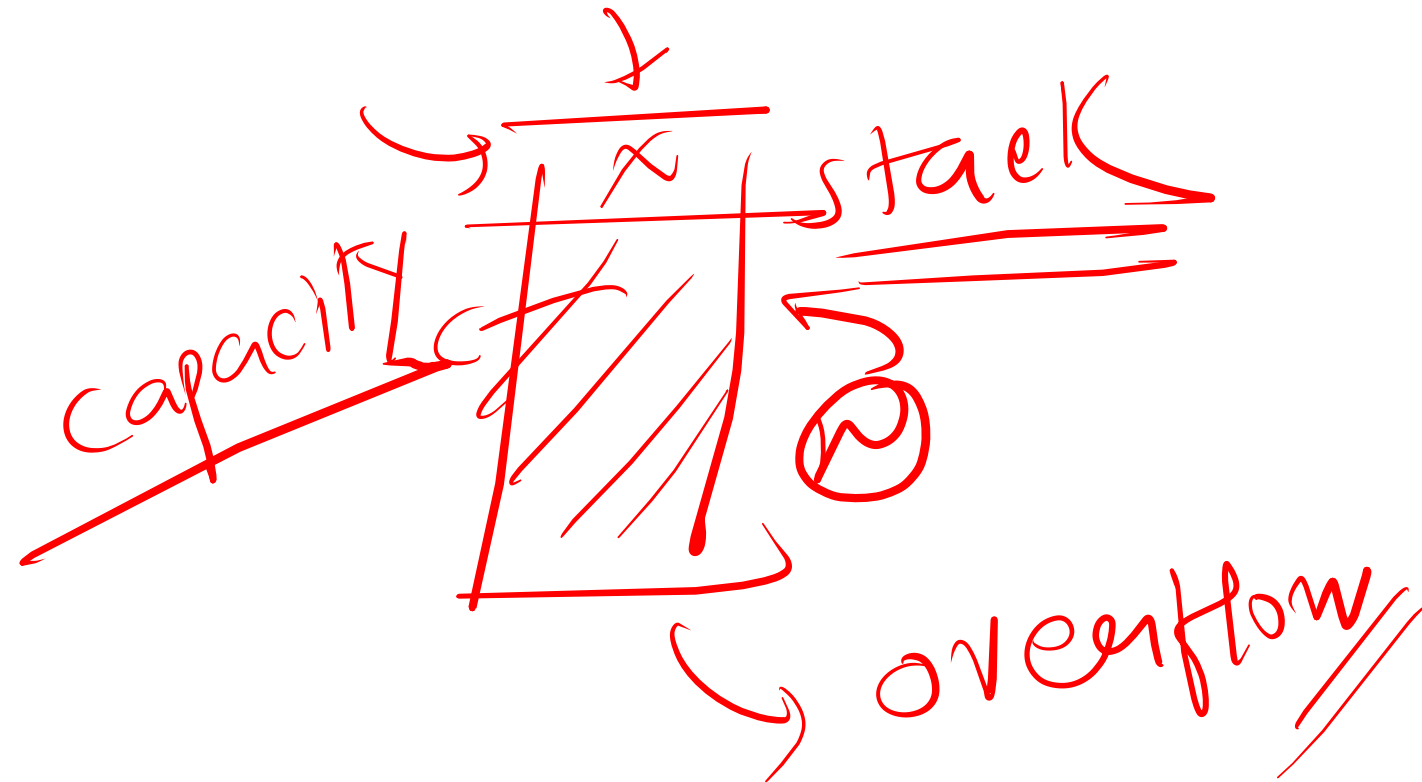


Stack

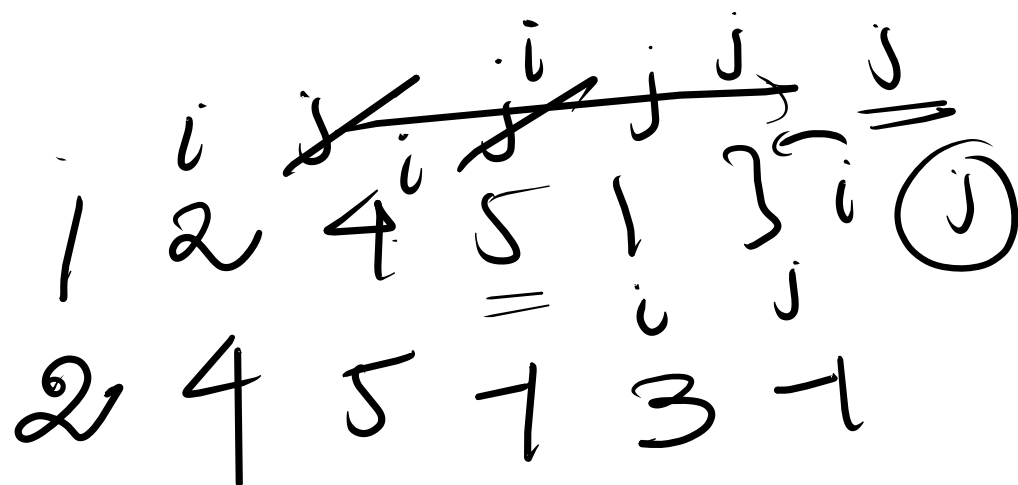
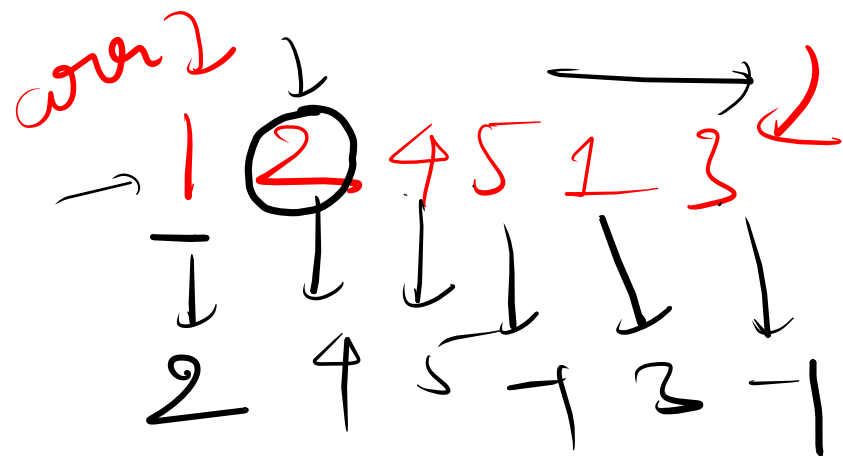
?







next greater element
↓
e



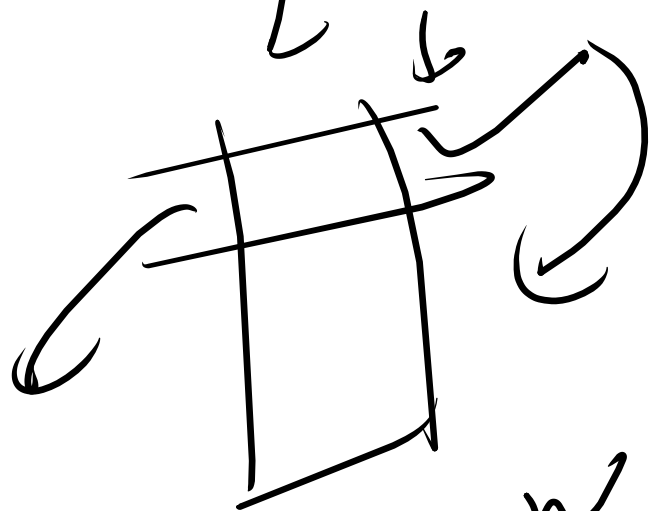
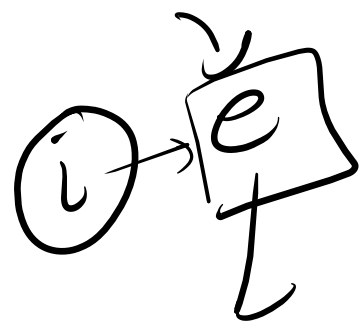
$$\textcircled{i} \quad \underline{\underline{j \rightarrow i+1 \dots n-1}}$$

$$i \rightarrow \textcircled{n} * \textcircled{n}$$

$$\underline{\underline{O(n^2)}}$$

$$\rightarrow O(1)$$

sc//

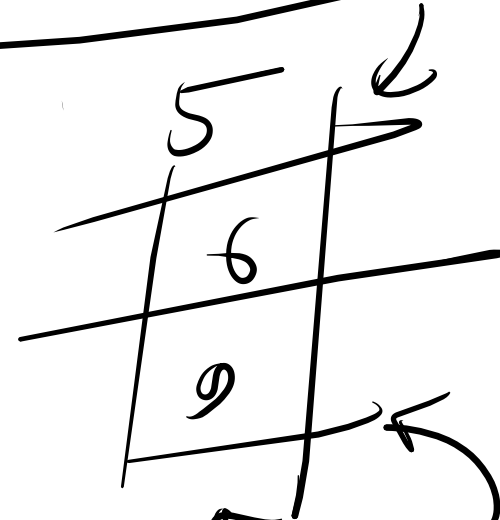
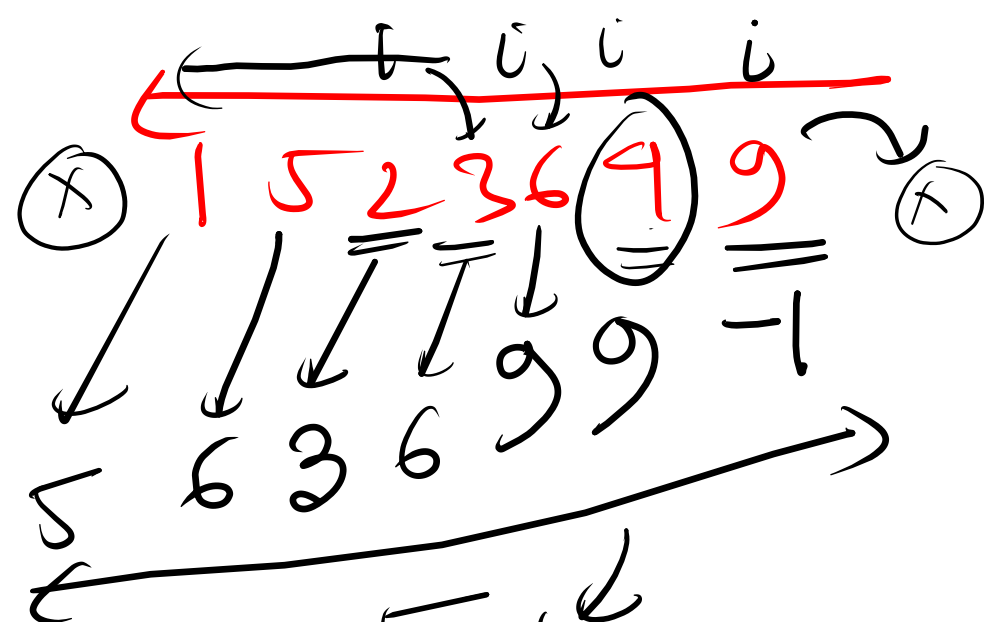


$2n + 2n$

$2n$

$O(\neq n)$

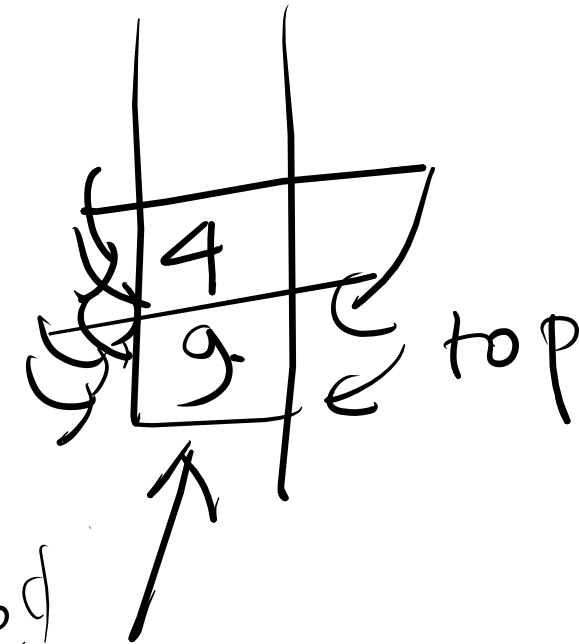
$O(n)$



introduced stack.

n

$O(n) \rightarrow SC$



extra
 $\downarrow$
 $n^2 \rightarrow \textcircled{2}$

extra space } $\textcircled{Tc}$

$sc \propto \frac{1}{Tc}$ $\downarrow$

$\textcircled{\times} \overline{7} \ 5 \ 8 \ 2 \ 1 \ 3$
 $\downarrow \ \downarrow \ \downarrow \ \downarrow \ \downarrow \ \downarrow$
 $8 \ 8 \ \textcircled{-1} \ 3 \ 3 \ \textcircled{-1}$
 $\downarrow \downarrow$
 $NG \rightarrow \textcircled{e}$

2
1
3

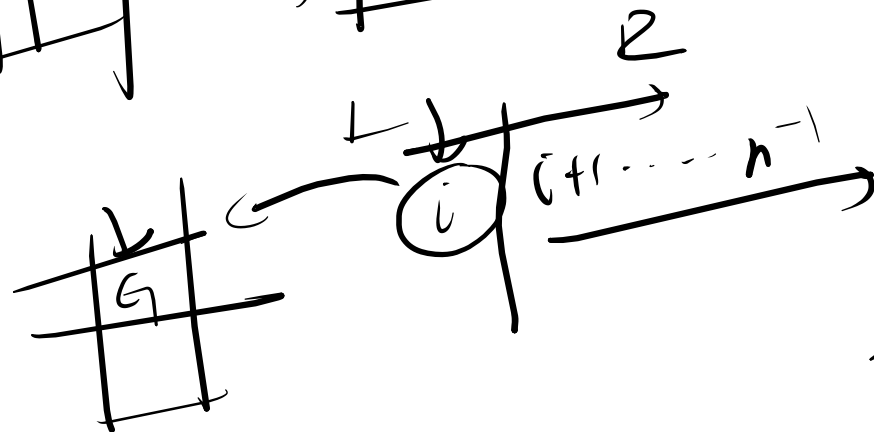
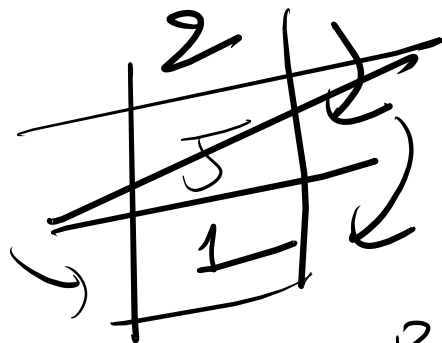
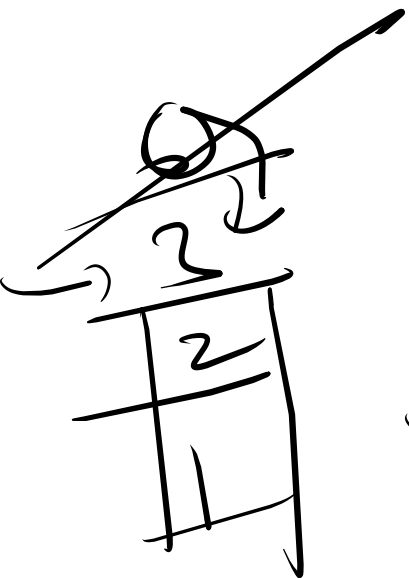
9	7
5	2
8	1


```

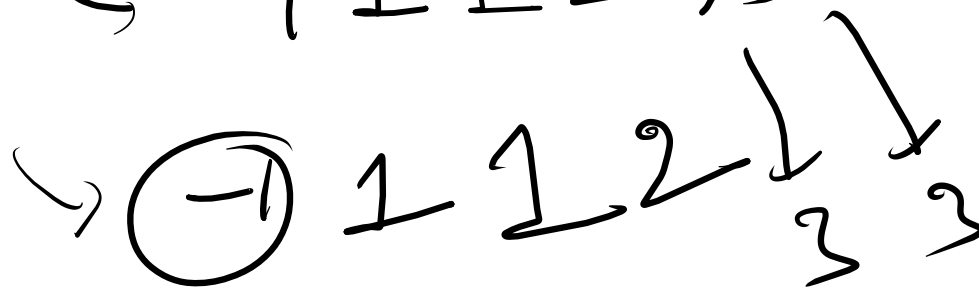
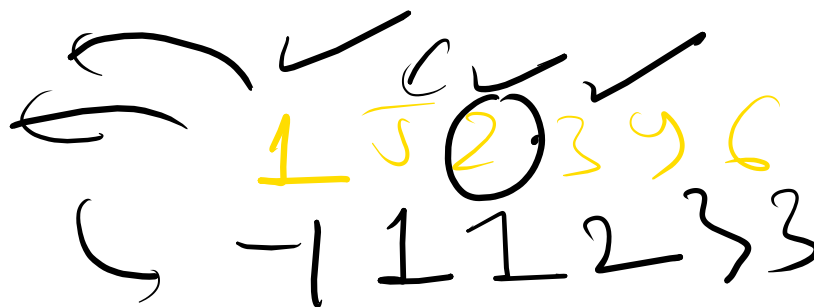
for(i=n-2; i>=0; i--) {
    while(!stk.empty() {
        arr[i] arr[i]
        >= stk.top()
        stk.pop();
    }
    if(stk.empty()) nge[i] = -1;
    else nge[i] = stk.top();
    stk.push(arr[i]);
}

```

stack <int> stk; arr[n];
 nge[n];
 nge[n-1] = -1;
 stk.push(arr[n-1]);

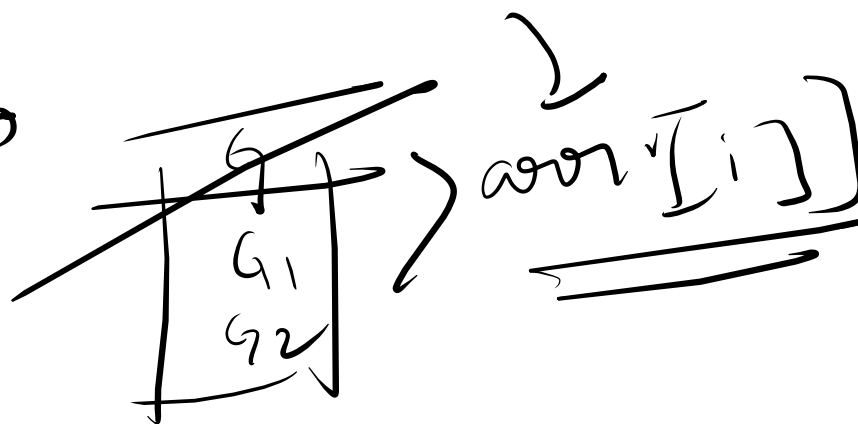


stack



→ PSE

→ e → array

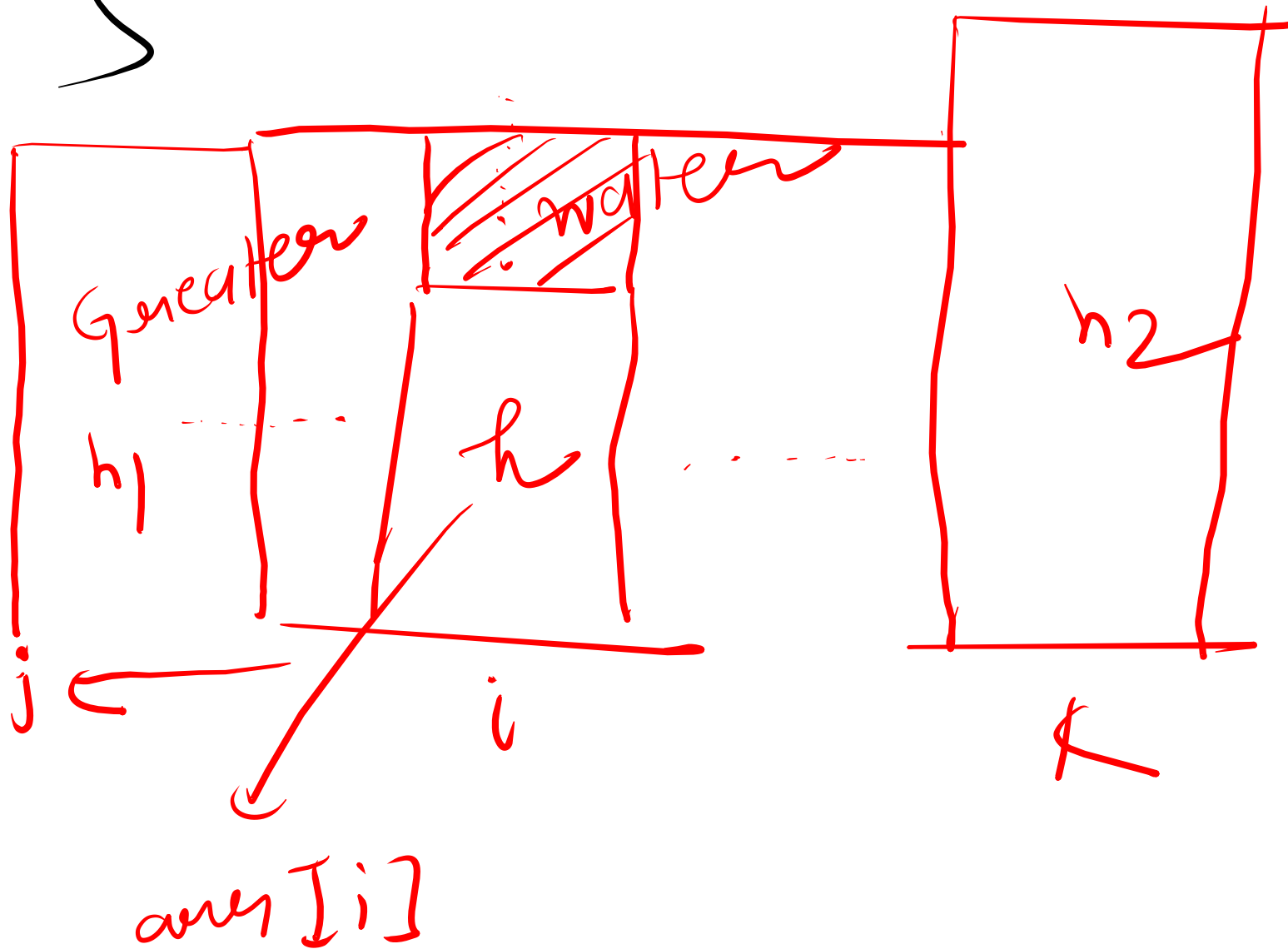


```
arr[n];  
pse[n];  
stack<int> stk;  
pse[0] = -1;  
stk.push(arr[0]);
```

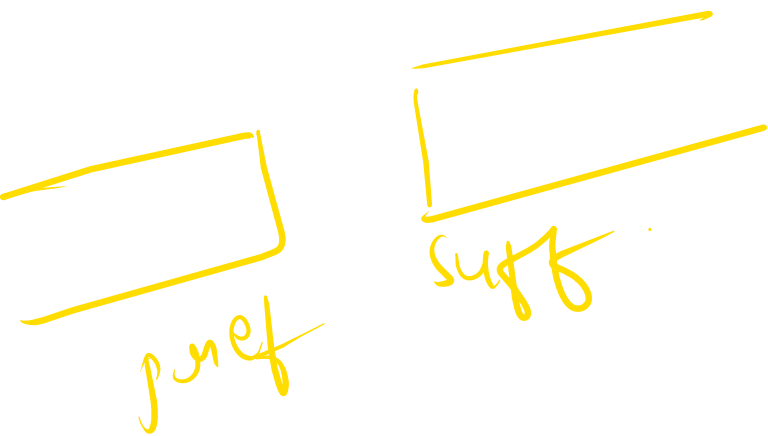
```
for (i = 1; i < n; i++) {  
    while (!stk.empty() &&  
           stk.top() > arr[i])  
        stk.pop();
```

```
    if (stk.empty())  
        pse[i] = -1;  
    else { pse[i] = stk.top(); }  
    stk.push(arr[i]);  
}
```

$$\frac{HSE}{PGE} \rightarrow \underline{\underline{HW}}$$

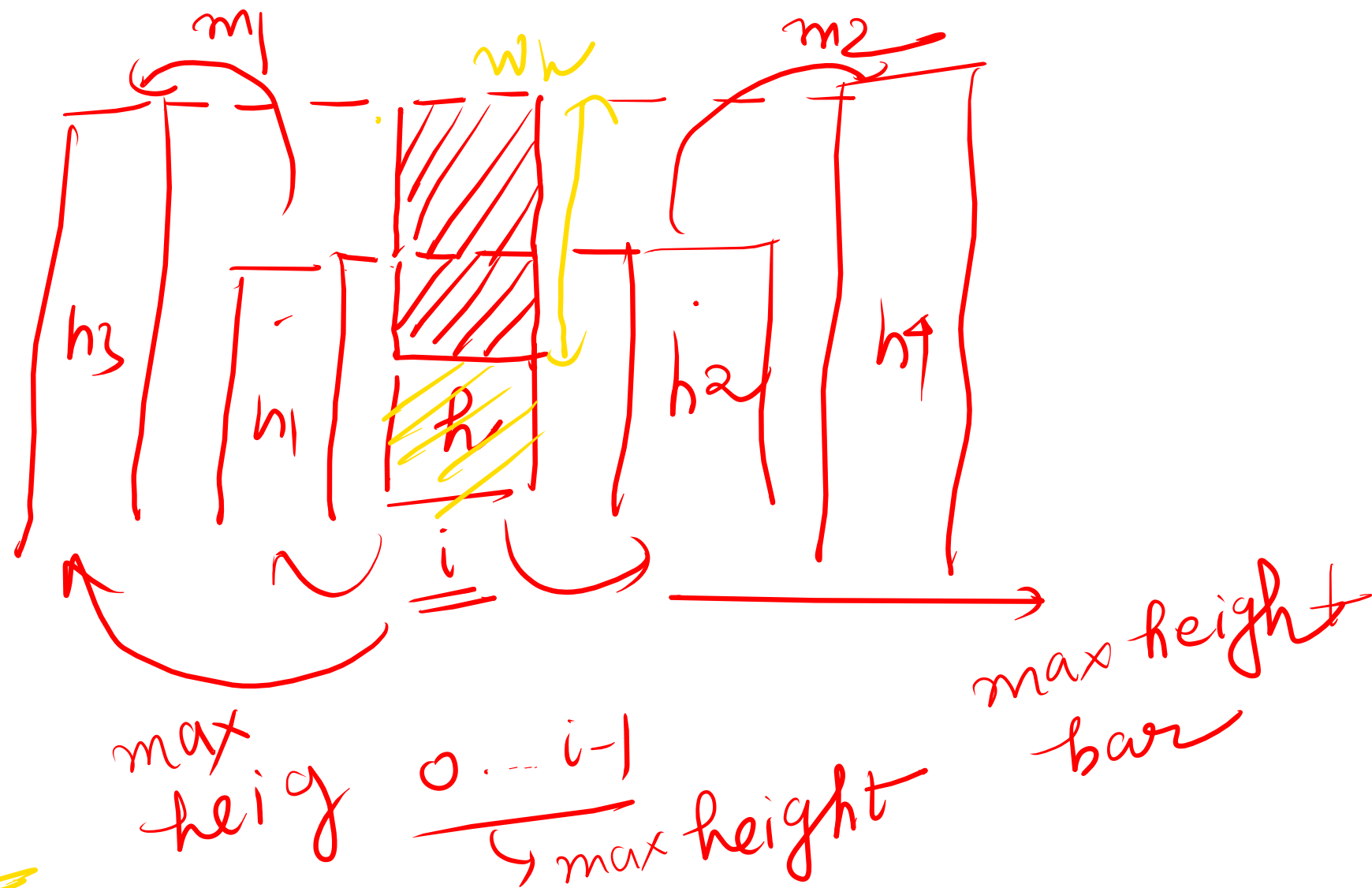


$$\min(m_1, m_2) - h$$



pref $\rightarrow$ pref[i] = 0 $\dots$ i

max



$(0 \dots n-1)$
 $(0 \dots i-1)$

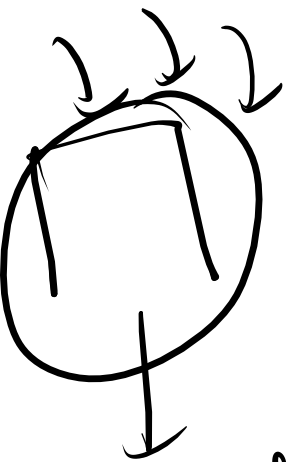
water[i] →
min(pref[i-1],
suff[i+1])
→ h[i];

$(0 \dots i)$
pref[i] = max(pref[i-1], h[i]);
 $\frac{0 \dots i-1}{\swarrow}$

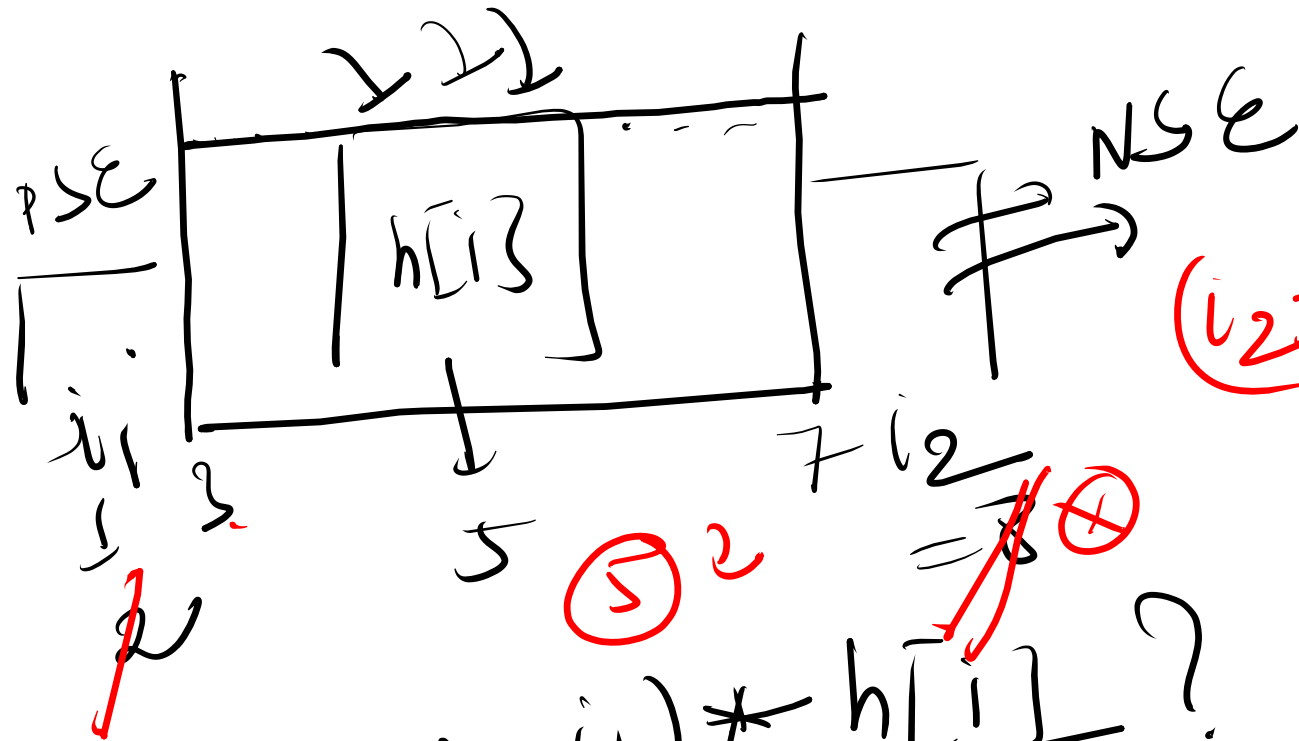
suff[i] = max(suff[i+1], h[i]);

$\frac{0 \dots \textcircled{n-1}}{\searrow}$
max?

$\textcircled{i} \frac{i+1 \dots n-1}{\searrow}$
max



height $\rightarrow$ rectangle



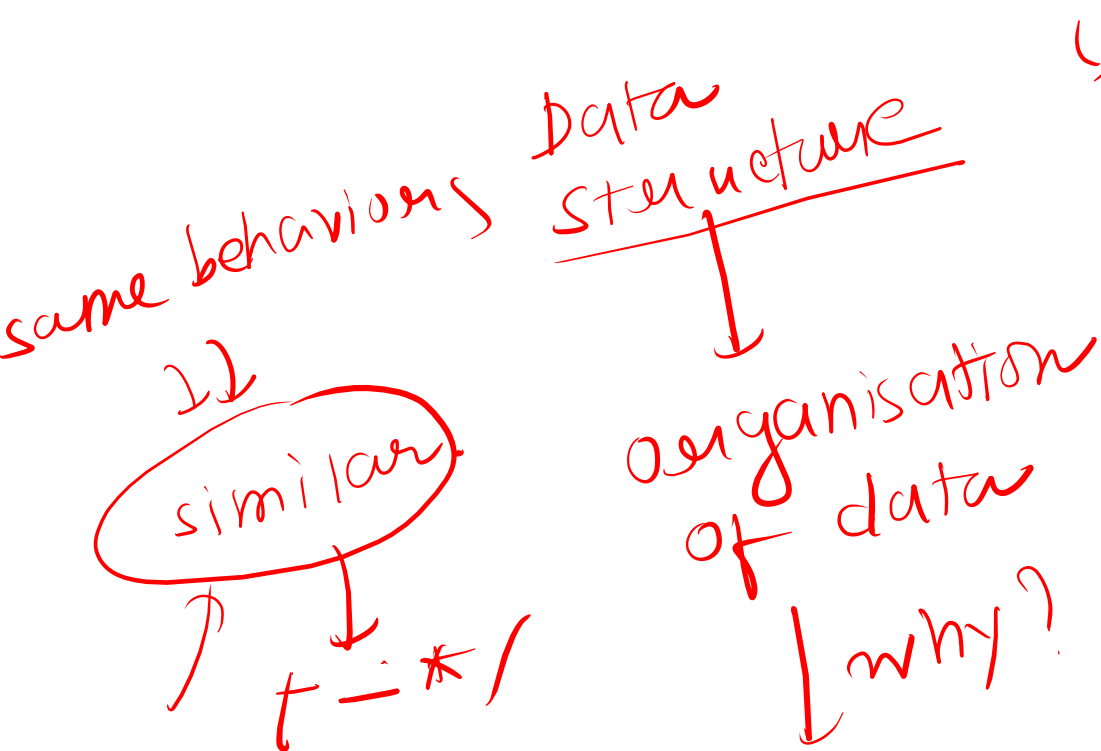
$$(i_2 - i_1 - 1)$$

$$5^2 = 25$$

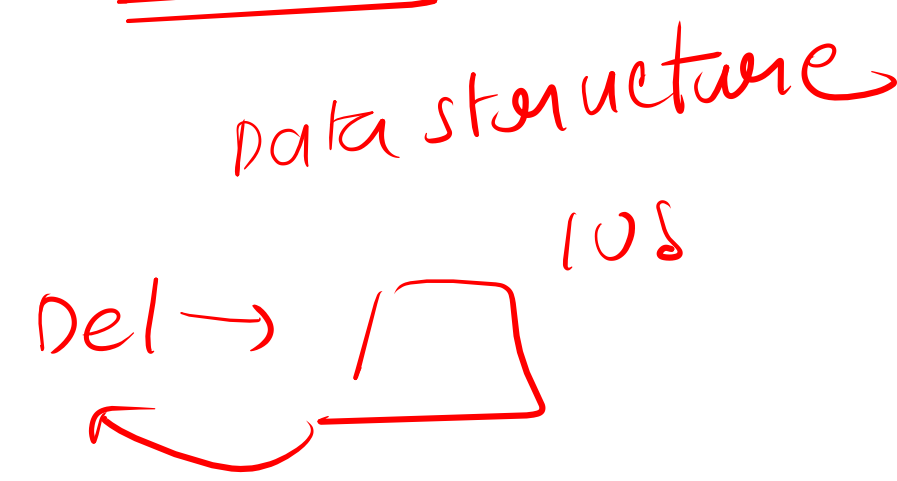
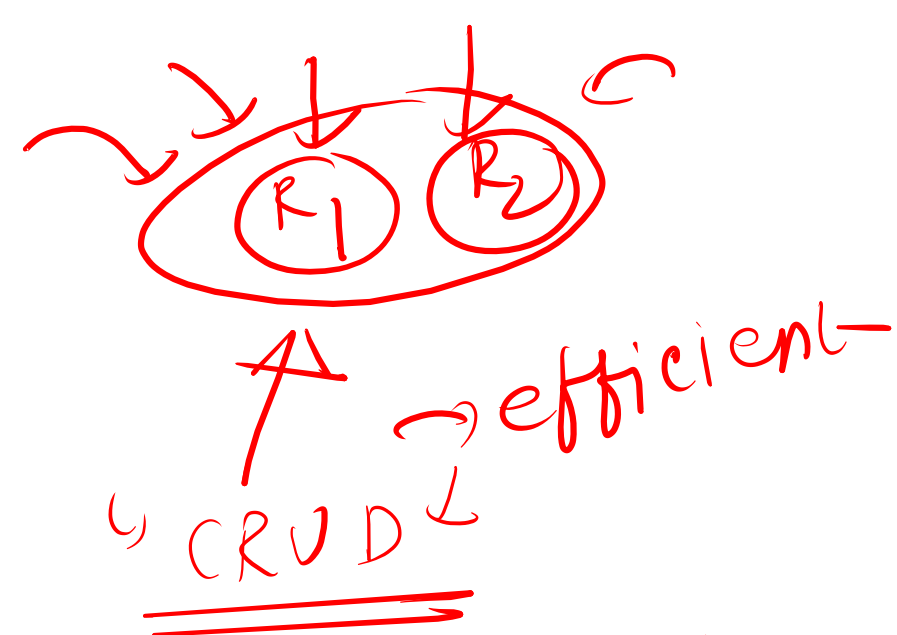
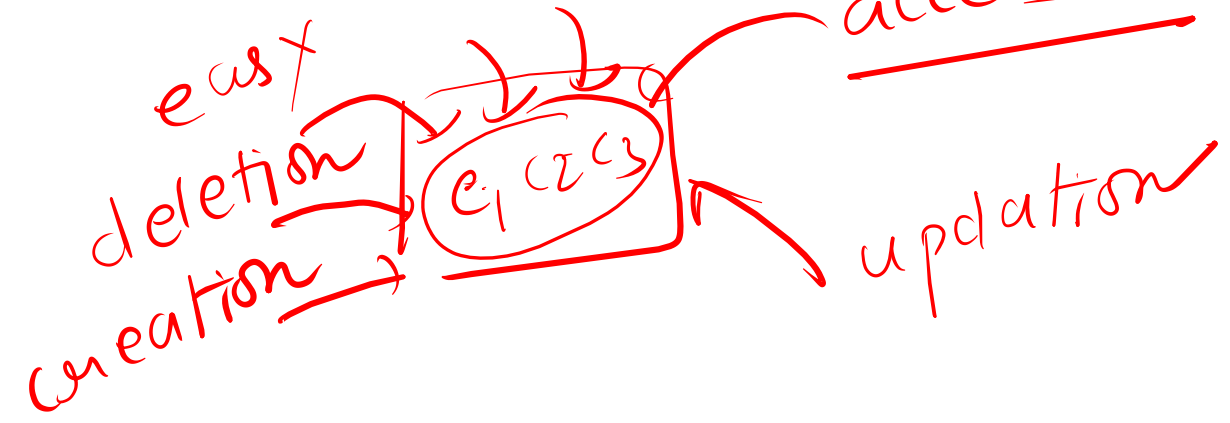
$$(i_2 - i_1) * \underline{h[i]} ?$$

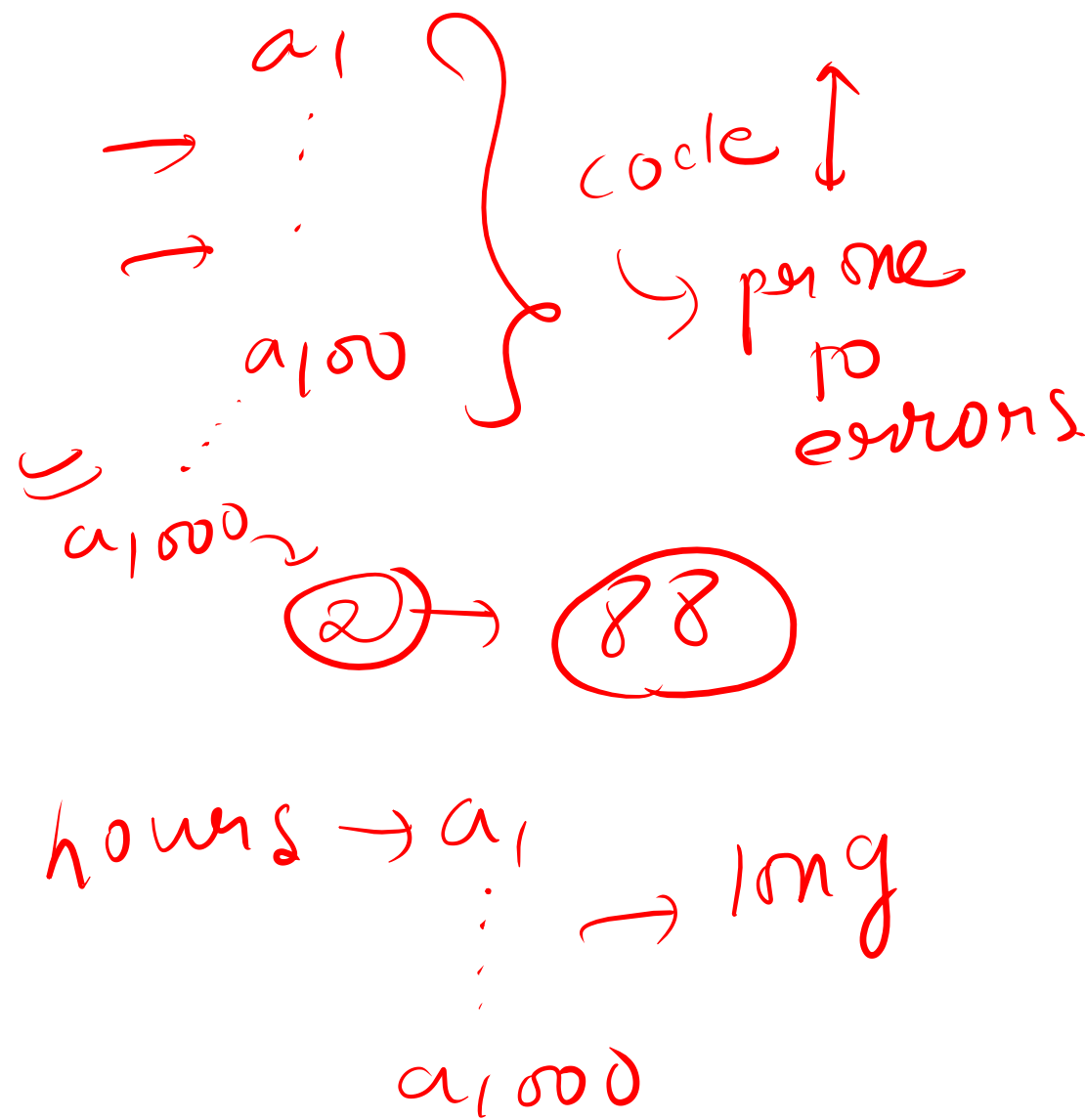
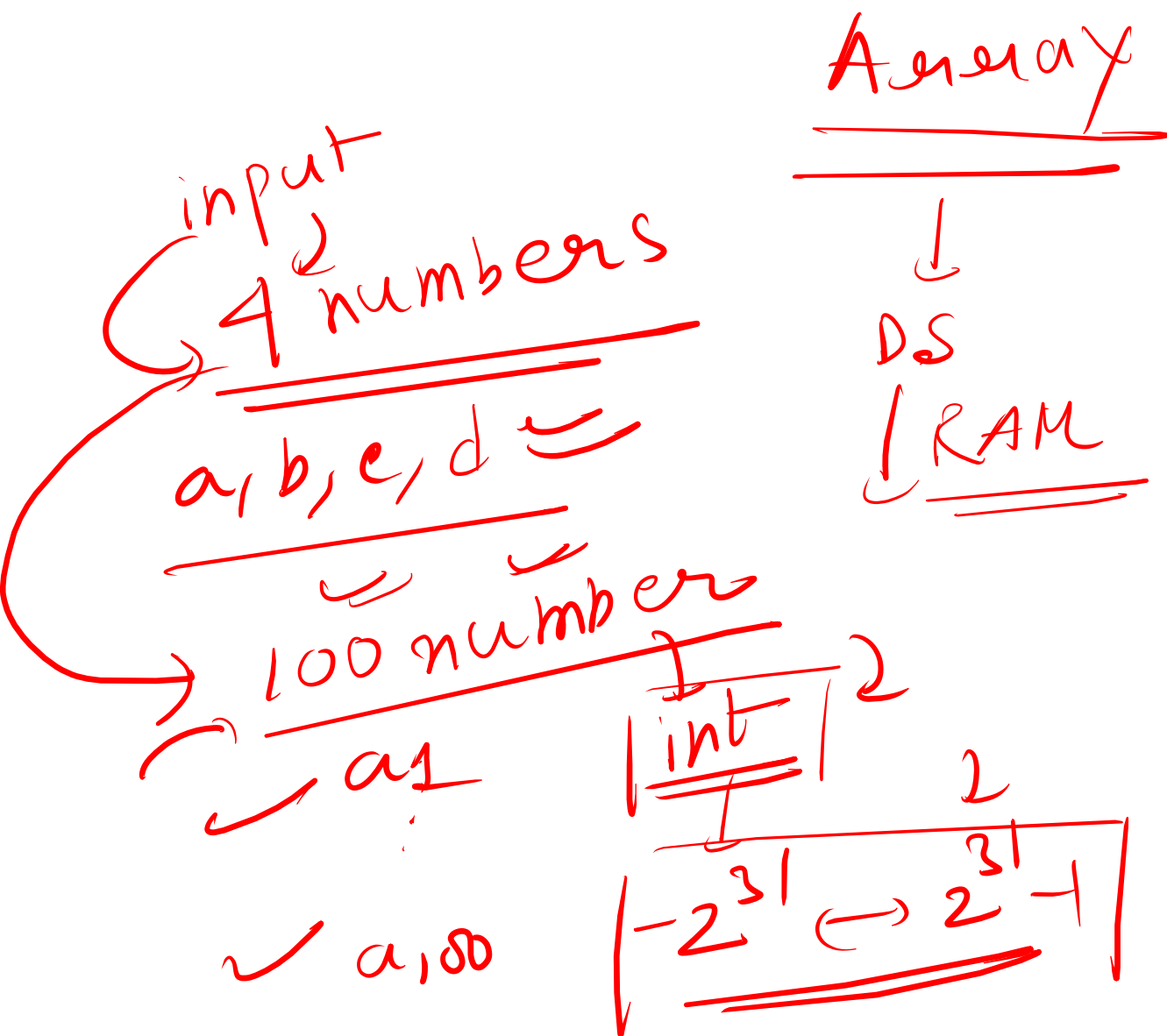
area

$$\underline{(i_2 - i_1) * h[i]} \rightarrow \text{area} \rightarrow \text{ans}$$



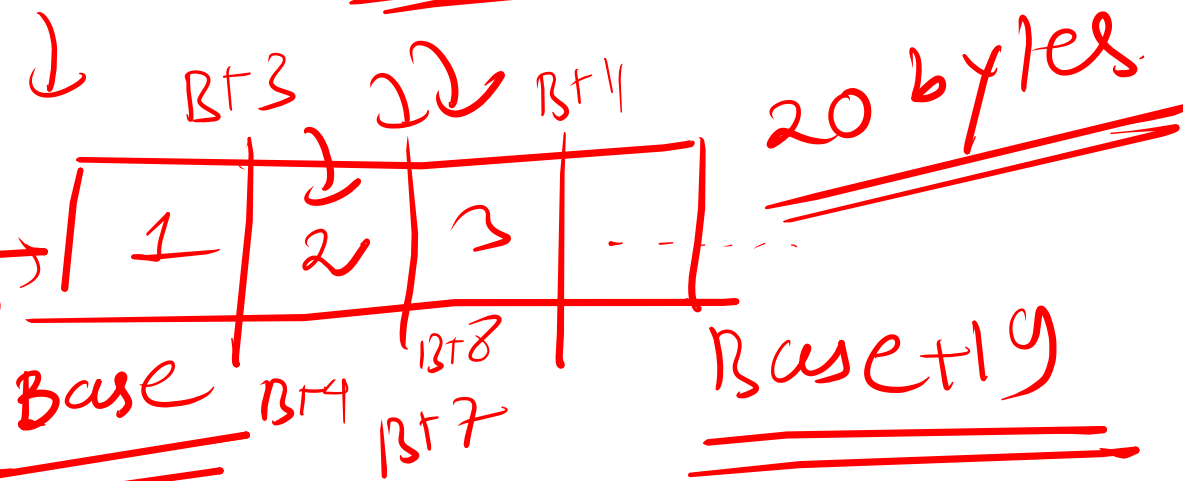
↳ Array ↓





`int arr[5];`

`arr` → 5 no. → 5 x 4 bytes?
int ↓



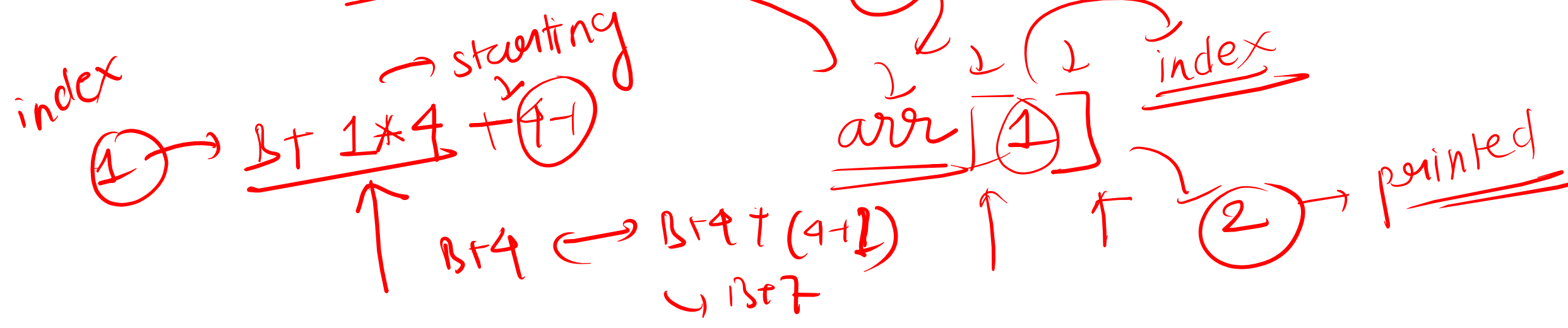
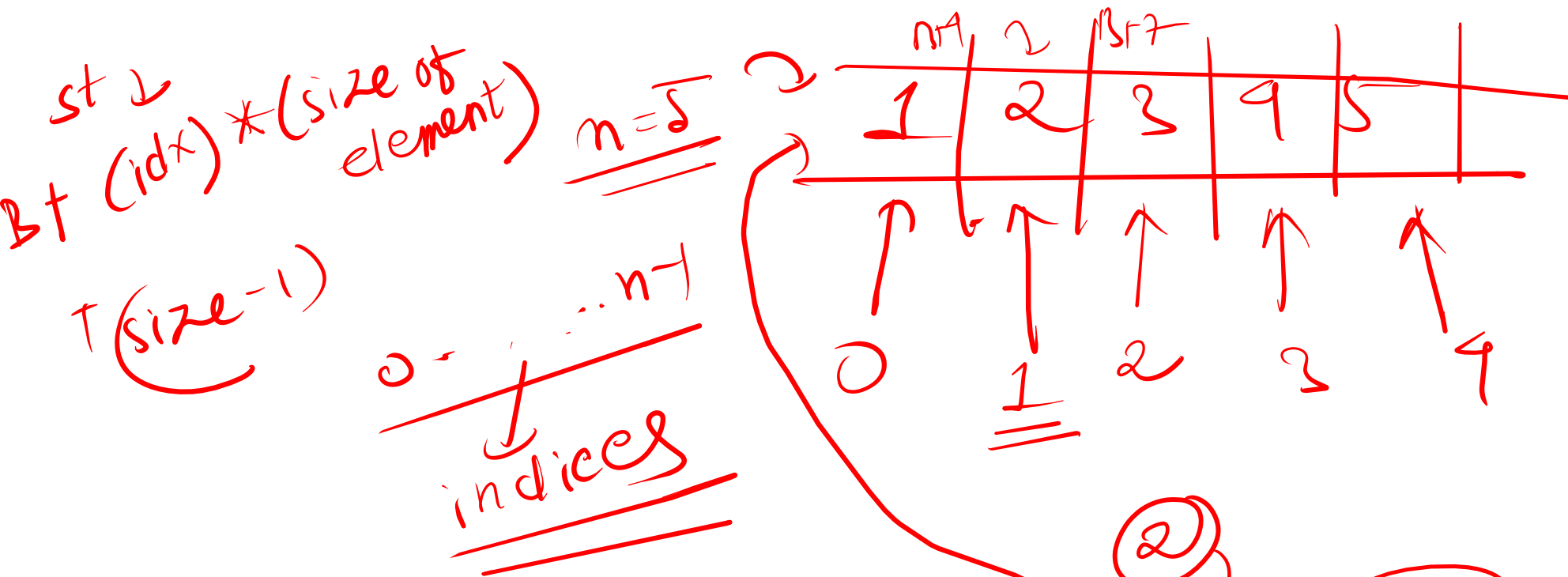
Base? → hexadecimal

② → B+4 ... B+7?

arr

`num` → 4 bytes.

B → B+3 → 1
B+4 ... B+7 → 2 ...

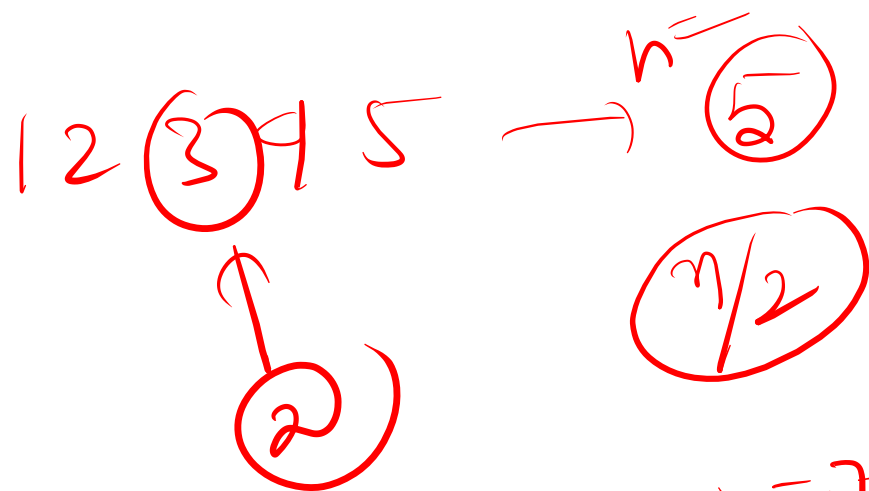
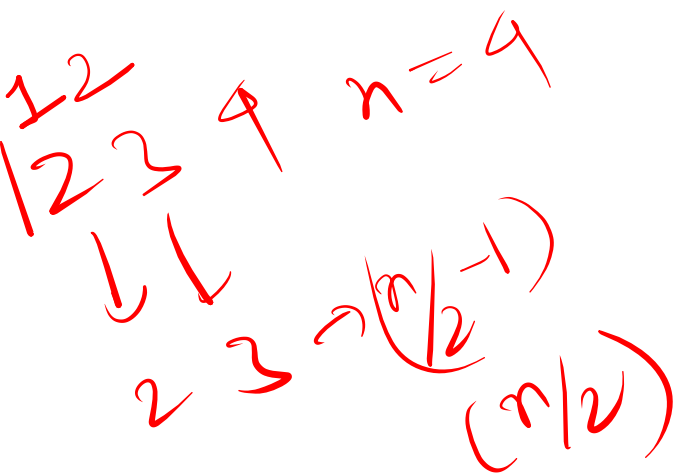


arr → 1 2 (3) 4 5 ↙ console
input.

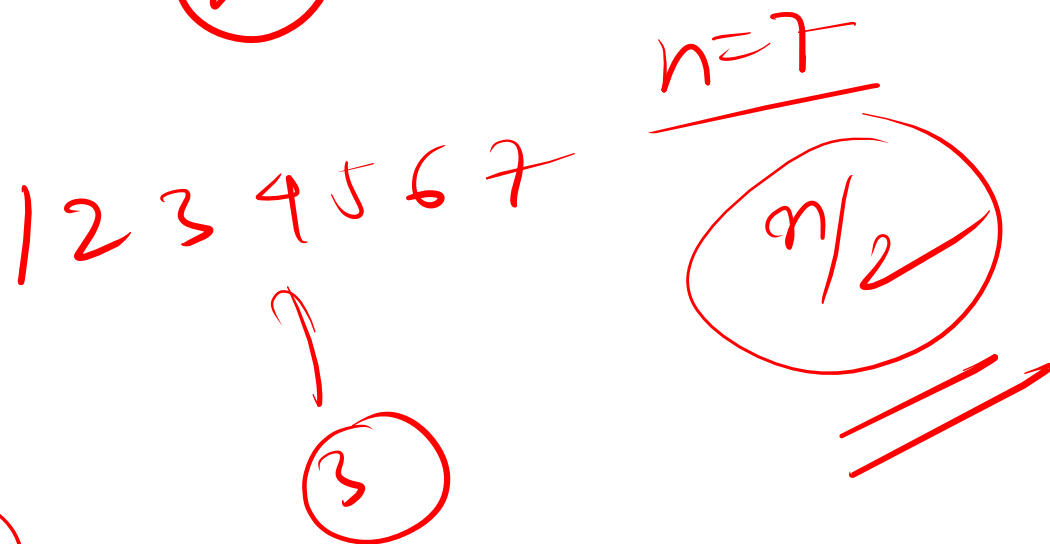
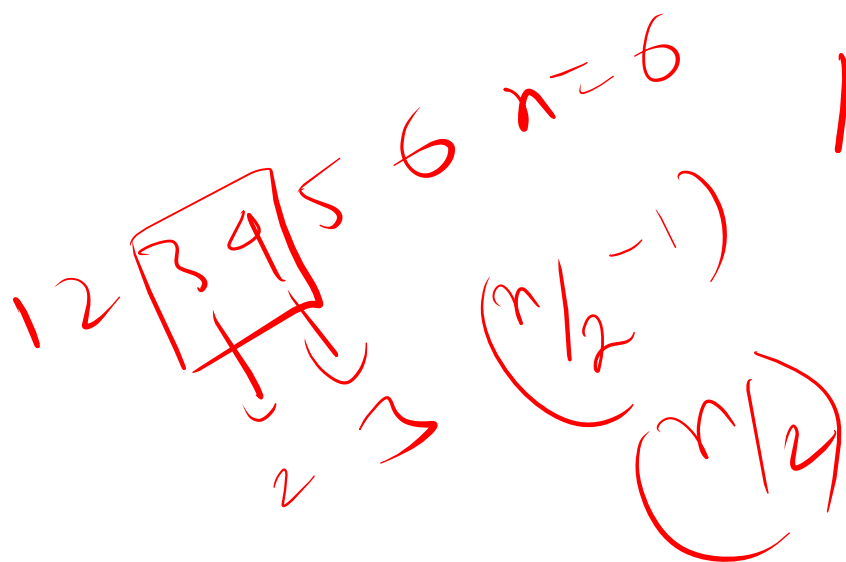
{ mid
element(s)

↓
(3) ✓

1 2 (3 4) 5 6
↓
3 4

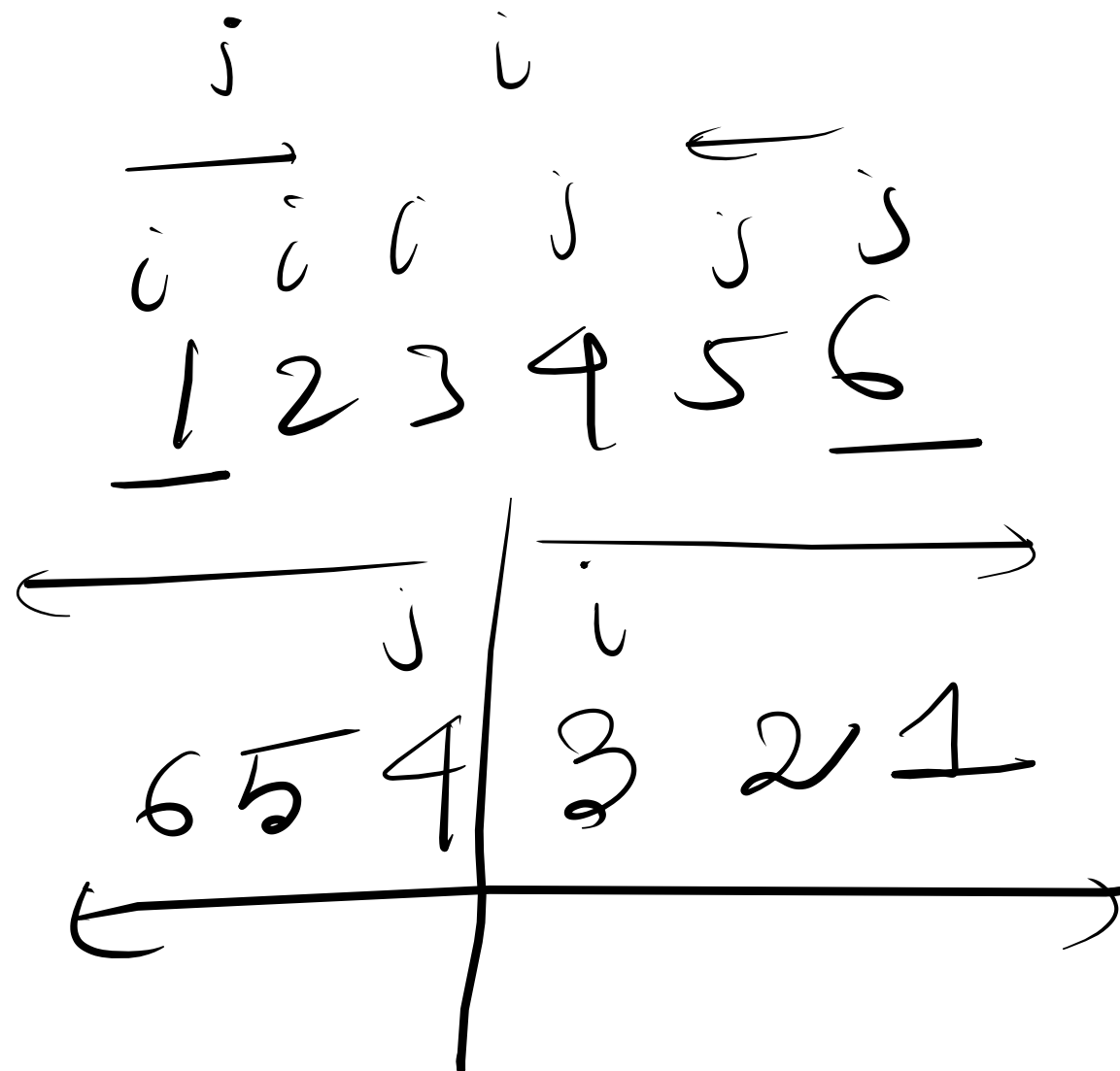


$\left. \begin{array}{l} \text{integer} \\ \text{div. ?} \end{array} \right\}$

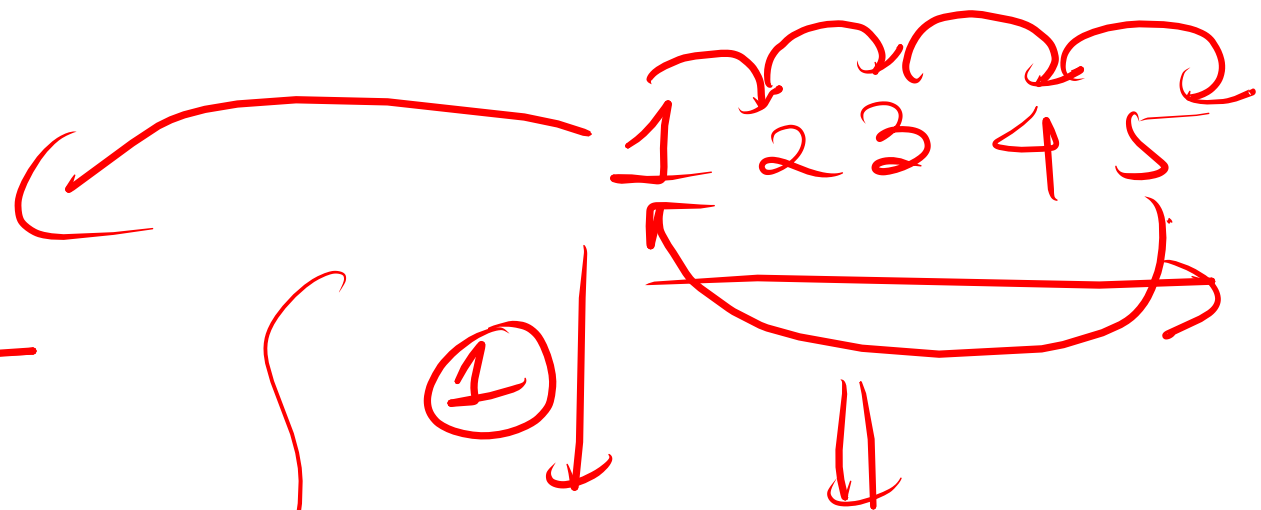




جاء



5 1 2 3 4

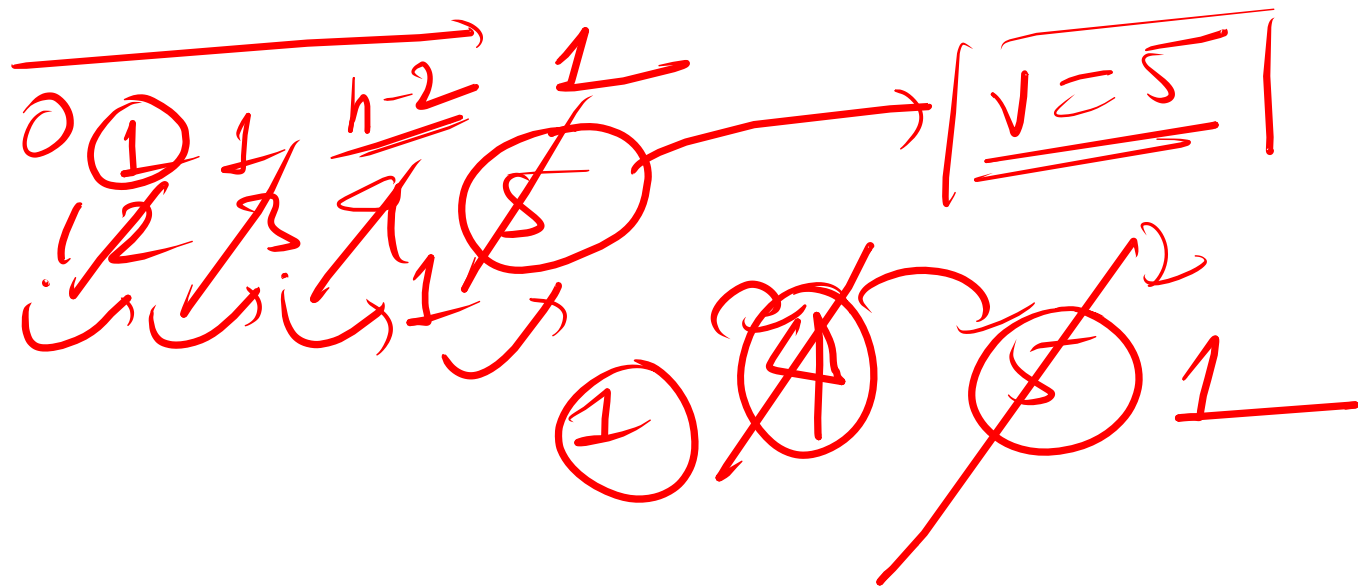


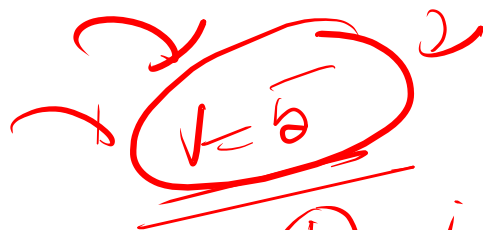
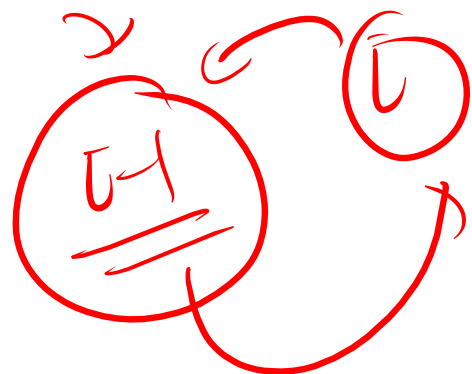
$n=5$

clockwise
direction
by one
step

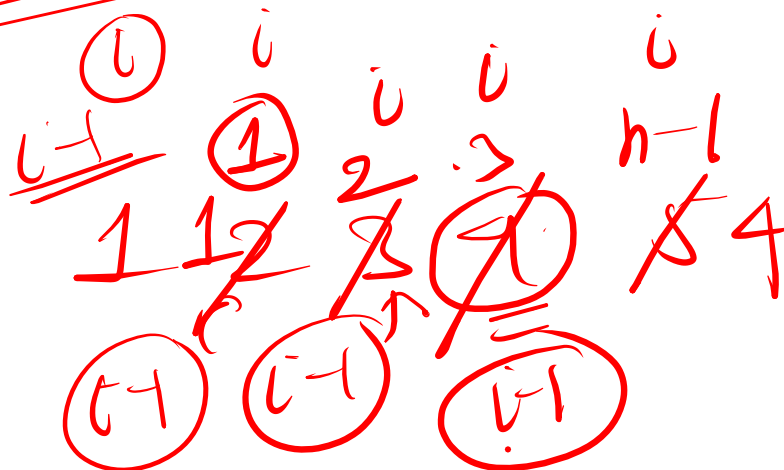
5 1 2 3 4

4
lost
3
lost
2
lost

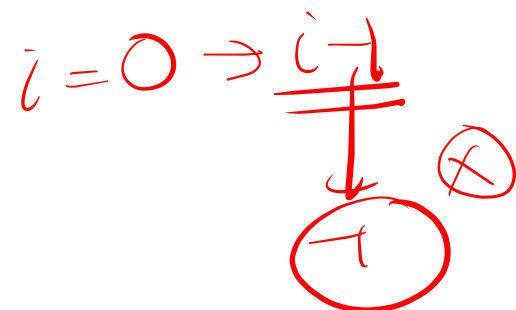




i

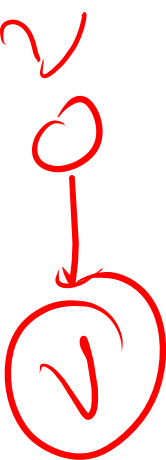


$i=0$ (X)



$n-2$ $\frac{n-1}{\downarrow}$ original $\frac{1}{\downarrow}$ lost (5)

5 1 2 3 4



i $i+1$ $i-1$ i

for ($i = n-1$; $i \geq 1$; $i--$)
 $arr[i] = arr[i-1];$

$v = arr[n-1]$

Rotated
arr.

$n-1$ 1 2 3 $\dots$ $n-1$

0 $\otimes$

$arr[0] = v;$